

WORLDS OF



SCIENCE FICTION

MAY 1963 • 40¢

TURNING POINT
by
POUL ANDERSON

THE GREEN WORLD

A Complete Novel of
Adventure Beyond The Stars

by **HAL CLEMENT**

DIE, SHADOW!
by
ALGIS BUDRYS



WORLDS OF



SCIENCE FICTION

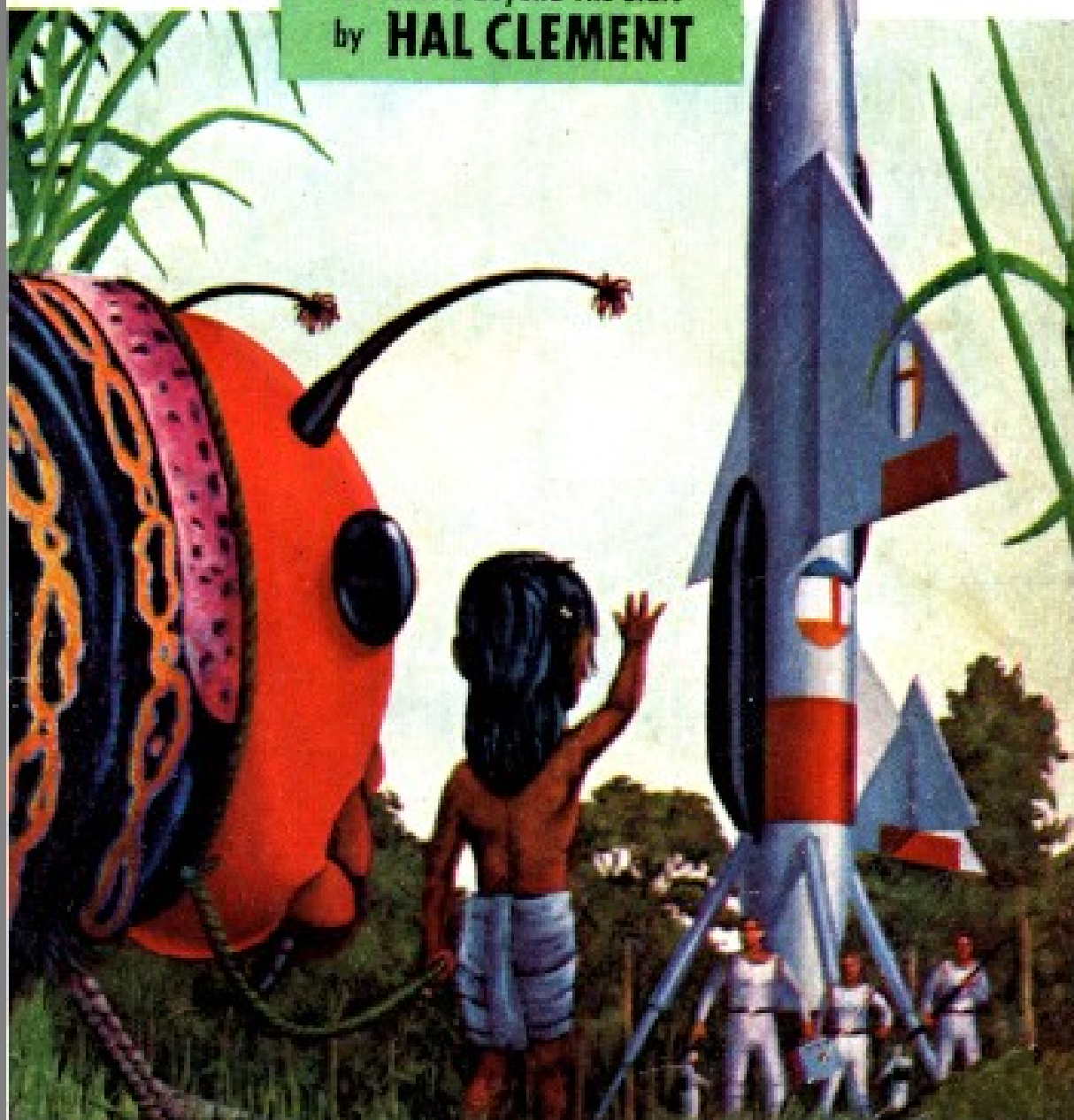
MAY 1963 • 40¢

TURNING POINT
by
POUL ANDERSON

THE GREEN WORLD

A Complete Novel of
Adventure Beyond The Stars
by **HAL CLEMENT**

DIE, SHADOW!
by
ALGIS BUDRYS



The Project Gutenberg eBook of The Green World

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Green World

Author: Hal Clement

Illustrator: John Giunta

Release date: February 25, 2020 [eBook #61509]

Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/61509

Credits: Produced by Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE GREEN
WORLD ***

THE GREEN WORLD

BY HAL CLEMENT

**The planet was an enigma—and
its solution was death!**

**[Transcriber's Note: This etext was produced from
Worlds of If Science Fiction, May 1963.
Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

I

A zoo can be a rather depressing place, or it can be a lot of fun, or it can be so dull as to make the mind wander elsewhere in self-defense. In fairness to Emeraude, Robin Lampert had to concede that this one was not quite in the last group. He had been able to keep his attention on the exhibits. This was, in a way, surprising; for while a frontier town has a perfect right to construct and maintain a zoo if it wishes, one can hardly expect such a place to do a very good job.

The present example was, it must be admitted, not too good. The exhibits were in fairly ordinary cages—barred for the larger creatures, glassed for the smaller ones. No particular attempt had been made to imitate natural surroundings. The place looked as artificial as bare concrete and iron could make it. To a person used to the luxuries provided their captive animals by the great cities of Earth and her sister planets, the environment might have been a gloomy one.

Lampert did not feel that way. He had no particular standards of what a zoo should be, and he would probably have considered attempts at reproduction of natural habitat a distracting waste of time. He was not a biologist, and had only one reason for visiting the Emeraude zoo; the guide had insisted upon it.

There was, of course, some justice in the demand. A man who was taking on the responsibility of caring for Lampert and his friends in the jungles of Viridis had a right to require that his charges know what they were facing. Lampert wanted to know, himself; so he had read conscientiously every placard on every cage he had been able to find. These had not been particularly informative, except in one or two cases. Most of the facts had been obvious from a look at the cages' inhabitants. Even a geophysicist could tell that the *Felodon*, for example, was carnivorous—after one of the creatures had bared a rather startling set of fangs by yawning in his face. The placard had told little more. Less, in fact, than McLaughlin had already said about the beasts.

On the other hand, it had been distinctly informative to read that a small, salamanderlike thing in one of the glass-fronted cages was as poisonous as

the most dangerous of Terrestrial snakes. There had been nothing in *its* appearance to betray the fact. It was at this point, in fact, that Lampert began really to awaken to what he was doing.

He was aroused all the way by McLaughlin's explanation of a number which appeared on a good many of the placards. Lampert had noticed it already. The number was always, it seemed, different, though always in the same place, and bore signs of much repainting. It bore no relationship to any classification scheme that Lampert knew, and neither of the paleontologists could enlighten him. Eventually he turned to McLaughlin and asked—not expecting a useful answer, since the man was a guide rather than a naturalist. However, the tall man gave a faint smile and replied without hesitation.

"That's just the number of human deaths known to have been caused by that animal this year." It did not comfort Lampert too greatly to learn that the year used was that of Viridis, some seventeen times as long as that of Earth. For the *Felodon* the number stood at twelve. This was not very much when compared to the annual losses from tigers in India during the nineteenth century. But this reflection was not particularly consoling. The human population of Viridis was so very small compared to that of India.

Lampert examined the creature thoughtfully. It was of moderate size as carnivores went—some four feet long without the tail—and looked rather harmless as long as it kept its mouth shut. It was lying in the center of the cage, so it was difficult to judge the length of its legs. It showed no trace of the tendency displayed by many captive animals, of lying against a wall or in a corner when relaxed; and there was none of the restless pacing so characteristic of Earth's big cats under similar circumstances. It simply lay and stared back at Lampert, so steadily that he never was sure whether or not the cold eyes were provided with lids.

"I never liked reptiles back home, but I think I like these creatures less." The voice of Mitsuitei, the little archaeologist, cut into Lampert's reverie.

"Don't let Hans or Ndomi hear you mention them in the same breath with reptiles," he answered.

"Well, I'm not fond of frogs, either."

"I'm afraid that wouldn't make them much happier. These are not even amphibians."

"They certainly are. I've been told that they lay eggs in water and have a tadpole stage—"

"I should have said they aren't Amphibians with a capital A. That is, they don't belong to the order Amphibia, since they are not genetically related to the corresponding order on Earth, as far as we know. Sulewayo gets quite peeved at people who try to lump terrestrial and extraterrestrial creatures in the same order. I believe that whoever decides things for biologists has decreed that on Viridis the dominant order is to be called Amphibids. It's a quibble, if you like. But I can see why they insist on it."

"Mph. So can I. Even now you sometimes run into people who go to great length to make you admit that there are pyramids both in Egypt and Mexico—and for that matter on Regulus Six—and infer from that that their makers had something in the way of common culture. I say these things are amphibians, without the Capital A, because they are at home both on land and in water. And a dictionary would back me up. I don't insist that they're related to those of Earth—any more than a Mayan pyramid has anything but geometry in common with an Egyptian one."

"But I've heard—"

"I'm sure you have, but it's a sore subject. I'll be open-minded if you like and admit that some Egyptian *may* have been blown across the Atlantic and taught architecture to the Americans, but I don't regard it as proved. What was that remark of yours—'as far as we know'—in connection with the ancestry of the amphibids? That's being at least as open-minded as I was, I would say."

"In a way, yes. I don't think anyone has seriously suggested that these things originated on Earth. However, a puzzle we're here to investigate still exists. How there could be life forms corresponding to those which took a good half billion years to evolve elsewhere, on a planet which by geophysical evidence hasn't been solid for forty million? *Someone* certainly has suggested that the world was stocked from outside. But certainly it hasn't been proved. I don't

think anyone has tried very hard, either. And I certainly won't, on a planet with as much radioactivity as this one."

"You think that would account for high-speed evolution?"

Lampert shrugged his shoulders, and began to stroll toward the next cage. "Ask the paleontologists. My opinion doesn't carry much weight."

Mitsuiei nodded, started to follow the geophysicist, and then turned back to stare once more at the carnivore lying a few feet away. It stared back unblinkingly.

The visit to the zoo was one of several, which continued until Lampert, Mitsuiei and the two paleontologists were able to identify each of a dozen animals which were most concerned in the death rate of Viridis. Apparently McLaughlin was not the only guide who did this. The zoo was equipped to give a "final examination" in which any creature the guide desired could be seen on a television screen from viewpoints quite different from those obtained in front of the cages. McLaughlin proved hard to satisfy.

Lampert did not blame him. He knew a lot about Viridis, of course. He had not only read of it in ordinary reference material, but had done much of the laboratory work on drill cores brought from the planet. His name had been one of those attached to the report giving the probable age of the planet's crust. At that time, however, the mental picture he held had been of continent distribution, rock strata, zones of diastrophic stress and the like. The question of the appearance, or even the existence, of plants, animals and people had simply never risen to conscious level in his mind.

That had changed, shortly before his arrival. The tramp spacer which had brought him and his group to Viridis had had to orbit about the world in free fall for several hours while its obsolete drive elements "cooled," and the passengers had examined the planet.

Lampert, oddly enough, had been as much impressed by the night side as by the sunlit hemisphere. The latter had shown, at twenty thousand kilometers, a fairly standard land and water pattern. The most unusual thing about it had been the almost perfect uniformity of the land coloration, a light green which bespoke, or at least implied, a virtually complete covering of vegetation.

By the time the ship had circled to the dark side, however, it was much closer to the surface; and Lampert would have expected to make out

luminous sparks and patches of towns and cities by the hundreds.

He saw just two, and was not really sure of those. For the rest, the planet was a vast, gray-black circle occulting a portion of the Milky Way. It was not absolutely black, either. Its contrast with the background of the galaxy was diminished by the glow in the upper atmosphere arising from the recombination of water molecules dissociated during the day by Beta Librae's fierce ultraviolet light. The center of the circle was darker than the edges, where the line of sight penetrated through more of the luminous gas.

But even this sight, unusual as it was, did not affect Lampert as much as the lack of city lights. He had done field work in lonely, wild places before, of course; but until now he had always had the feeling of being in an island of wilderness more or less surrounded by civilization. On Viridis it was the civilized spots which formed the islands. And very small islands they were. There was no known native intelligent race, and settlements of alien races such as the men from Earth were still few and far between.

So Lampert was prepared for McLaughlin's care in readying the group for its trip. He was even glad of it, though he would probably not have admitted to being at all afraid of the venture. He would simply have said that it was nice to have a guide who took his responsibilities seriously.

That of course, did not mean that Lampert was intending to disavow any of his own responsibilities. He, like McLaughlin, had been keeping a careful eye on the other members of the group, looking for the signs of impatience or ill temper which could be the seeds of serious trouble if the journey were prolonged. He had come to tentative conclusions about this during the flight from Earth, but was pleased to see that, apparently, men who could stand the enforced companionship of a tramp spacer were also able to retain their senses of humor in the steam-bath environment of Viridis.

Sulewayo, of course, had seemed safe from the first. A man who has spent his formative years in the Congo rain forests where his ancestors had lived for generations was ideal for this world. His sense of humor was extremely durable. Lampert suspected that it might sometimes be a little too good. Mitsuitei, the archaeologist, had once or twice appeared to resent some of

the young fellow's remarks, though not to an extent where Lampert had felt the need for introducing his own personality into the matter. Krendall, nearly twice Sulewayo's age, seemed to be a check on the younger man anyway; he was a member of the same profession, and Sulewayo would have been the first to admit his respect for Krendall's work in the field. Under the circumstances, Lampert felt that the group was well matched.

Whether it would be able to do the job it had undertaken was another matter. The news reports had spoken glibly of the expedition which was going to "solve the mysteries of Viridis once and for all." Lampert, like any other scientist, knew perfectly well that the solution of the present crop of mysteries about the planet would almost certainly be achieved only at the cost of creating an even greater number of new ones. Even the guide, who was admittedly no scientist, had expressed a similar opinion, though his was based on a general pessimism bred of familiarity with the planet. However, he had undertaken to get them to the sort of country they wanted; and from then on the problem solving was not his affair.

The scientists, whatever may have been their feeling about matters of personal safety, were eager to start, which tended to cause rapid progress in McLaughlin's animal recognition school. Another factor tending toward the same result was that there was little in Emeraude for such men to do, except learn. The town was still small. It had a spaceport and airport, which furnished little entertainment, docks which could amuse for a while but not indefinitely, and warehouses which were completely uninteresting to geologists, paleontologists and archaeologists. There was no museum. The numerous specimens of mineral, animal, and vegetable matter collected on the planet invariably wound up on outbound spacecraft. The zoo, which the town maintained for purely practical reasons, was about the only thing that was left.

In consequence, not many days passed before all four scientists were able to meet McLaughlin's requirements. Sulewayo was annoyed by the guide's addition of a short postgraduate course in edible flora and fauna, but admitted that the knowledge might well be useful. However, he made no secret of his satisfaction when McLaughlin finally announced that, as far as he was concerned, the journey could begin at any time.

All four rechecked their equipment—that of Lampert was by far the bulkiest—and, everyone satisfied with the group's ability both to live and to work in

the steam bath that was the world of Viridis, they watched the harbor on which Emeraude was located shrink and blend into the rest of the shoreline behind them. Within a few minutes only the restless surface of Green Bay was visible through the ever-present haze....

The jaws of the Felodon abruptly stopped moving and its forelegs straightened, bringing the fanged head up and away from the kill it had just made.

If a man had been there he would neither have seen nor heard the disturbing factor, for a thunderstorm a few miles to the west was emitting an almost continuous growl and the towering trees shut out nearly all the sky. Nevertheless the beast appeared to sense something out of the ordinary. It twisted its short but supple neck ceaselessly, rocking its head from side to side to bring first one eye and tympanic membrane to bear on the jungle roof, then the other. Sometimes it froze motionless for a long moment, and a watcher would have sworn that its minute brain was struggling with a thought. If this were the case, the thought must have been both unusual and unpleasant, for under normal circumstances nothing short of overwhelming force would have driven a Felodon from its meal. Now, however, the hind legs slowly straightened and the creature came erect. For another moment it stood motionless, took a step or two away from the body, and stopped again.

Abruptly, as though in defiance of some impulse, it turned back, lowered the murderously armed head and tore a huge mouthful of flesh from the carcass. Then, like a child leaving the cookie jar as its mother approaches, it leaped away into the underbrush, still swallowing.

Its speed was high and it did not have far to go. The jungle thinned in a few hundred meters to the point where some sky became visible, and a short distance further the riotous plant growth vanished completely to give place to an open beach. Here the creature stopped and repeated its search of the hemisphere overhead.

This time it found what it sought.

Along the line of the beach, perhaps a kilometer out to sea, the thing came flying. It must have been utterly different from anything the Felodon could

ever have seen, but no sign of fear appeared in the beast's demeanor. It stood on the beach, well away from the shelter of the jungle and certainly in full view from above, its head following the flying object and a fearful snarl—which might or might not have been its normal expression of hunger—giving its face an almost mammalian cast.

This thing was larger by far than any flying creature the Felodon knew—incomparably larger than the Felodon itself. Its details were hard to make out through the hazy air, and would have meant little to the flesh-eater in any case. The most noticeable characteristic was the steady, whistling hum that proceeded from it. There was a suggestion of motion, too, which might have been wings or might not. Actually, the thing was little more than a dark dot against the purplish-blue sky. At the moment no sunlight was striking it directly, for it was in the shadow of the thunderhead. Perhaps this prevented the animal below from being bothered by another unusual feature it possessed, though even the appearance of this last characteristic produced no sign of fear when it finally came. This occurred shortly after the flying thing passed, while it was still quite close. It moved out of the shadow of the great cloud and, as the greenish sunlight struck it, the eyes of the watching creature were dazzled by a gleam of metal.

This was certainly something it had never seen, for native metal on Viridis is just about as common as it was on Earth before men began to pry it out of its ores. Viridis has an oxygen-rich atmosphere and plenty of moisture, and pure aluminum or chromium just doesn't occur in that environment.

Strange or not, however, the gleam did not appear to affect the Felodon's rudimentary sense of fear. For just an instant it paused as the flying thing hummed on into the northeast; just once it looked back toward the point in the jungle where it had left its kill—a point from which eloquent sounds were now coming, betraying the presence of carrion-eaters; just one step it took in that direction. Then it turned away as abruptly as it had from the meal a few minutes before. With the same purposeful air it had displayed on the way out of the jungle it headed down the beach in the direction taken by the flying piece of metal.

Though the animal's speed was high, the humming soon faded out ahead of it.



However, this did not seem to cause any inconvenience; the Felodon moved on, with a gait that might have been called a fast walk or a slow run, never hesitating, never pausing. It remained silent. Smaller creatures which might have given it a wide berth had they heard the hunting call now sprang away almost from underfoot. It paid them no heed, but continued on its way while the green sun settled into the jungle behind and to its left. The fact that its recent kill was now little more than a skeleton did not seem to bother it. Perhaps it had forgotten.

II

The humming was a little more noticeable in the helicopter cabin, but not much. John McLaughlin, sprawled as comfortably as his two meters of height would permit in its confines, had noticed the sound only at first; and after remarking to himself that they seemed to be building better ion turbines since he had left Earth, had permitted his thoughts to wander in other directions. These did not concern Felodons; the interest there was not, at the moment, mutual. The rather crowded cabin offered material enough for consideration.

McLaughlin was not a scientist by training, but neither was he the sort of guide that might have been found in Yukon or Amazon territory a few centuries back. He did not despise people merely because they were, by his

standards, greenhorns. He knew that each of the other men now sharing this cabin with him was an expert in his own field, even though none of them, in spite of his training, would have been able to survive for more than a day in the jungles of Viridis. After all, why should they have learned such an art? There were other things worth learning, and one could always hire McLaughlin if a need to visit the jungles developed. Since this particular party had done just that, they were evidently a fairly practical crew.

They were not talking very much, which from the guide's viewpoint was an additional point in their favor. They already knew what they planned to do, and saw no point in repeating what had already been said. Of course, if they should fail to find the area they were seeking, there would be talk—all of it aimed at McLaughlin; but he had no fear on that score. There were few enough mountains on Viridis, and of those few by far the greater number were volcanic cinder cones. When these scientists had specified a region of tilted-block or folded mountains, the guide had been more than dubious at first. It had taken him time to recall that there was a small area meeting these specifications less than fifteen hundred miles from the spaceport at Emeraude. He was not himself a geologist, but pictures and diagrams had been used freely in explaining to him just what was wanted, and he was quite certain that the party would be satisfied with what he had to offer.

A slight rocking in the hitherto steady motion of the helicopter roused him from this line of reverie. They were already several hours from Emeraude, and McLaughlin realized that he should have been paying more attention to the course. He straightened up in his seat and looked out.

To the left and ahead was a huge thunderhead, whose satellite air currents had probably caused the variation on the helicopter's flight path. More important, there was land in sight. McLaughlin knew that the long flight across Green Bay was over. He waited, however, before saying anything. He had given the pilot full instructions as to the route before take-off, and he wanted to see whether those had been clear enough.

Apparently they had. Without asking questions or even looking back at the guide, Lampert swung the aircraft from its northerly heading onto one which paralleled the shoreline, a turn of about forty-five degrees to the right, and the helicopter resumed its steady flight.

McLaughlin did not relax. From now on the route was a little more difficult to follow, and there were not too many more hours of daylight. The shadowless night glow which made vision relatively easy after sunset did not lend itself to aerial navigation over a very poorly mapped world. He kept his eyes on the shoreline, watching for the landmarks he had not seen for many months—and then not from above. He did not see the Felodon which became so intensely interested in the helicopter. If he had, he would have attached little importance to the creature's presence, and he could not possibly have seen its actions in sufficient detail to catch any peculiarities in them.

No one else saw the beast, either. The change in course had roused most of the party from whatever lines of thought they had been pursuing, as it had McLaughlin, and most of them were looking out the windows; but they were interested in what lay ahead, not below. Sometime soon the relative monotony of jungle and swamp should be relieved by rising ground, indicating the nearness of the mountains they sought; and the helicopter's flight altitude of some two thousand feet was low enough to permit any significant rise of terrain to be visible. Sulewayo, the younger paleontologist, made a remark to that effect, which passed without comment. Real conversation did not start for some minutes.

"As I understand it, we have one more course change before we see the mountains. Isn't there a river we have to follow for a time, String?" Lampert asked the question without looking back.

"That's right," McLaughlin replied. "It runs into Green Bay from almost straight north, and about a hundred miles inland makes a turn to the east. That's general direction. It winds a lot."

"It would, in country as nearly peneplaned as this," muttered Lampert under his breath.

"The mountains you want start about sixty airline miles from the big bend. If you trust your gyro compass enough, you can head for them directly from the river mouth. If you have any doubt about being able to hold a line, though, follow the river. I doubt that there are any good landmarks

otherwise. Of course, I've only seen the area from the surface and close to the river, but I'd be very surprised if there was anything around but the swamp-and-jungle mess we're over now."

"So would I. We'll stay in sight of the river, but edge as far east as visibility lets us." The guide approved this plan with a nod, and the conversation lapsed for several minutes. The silence was finally broken once more by Sulewayo.

"I hope these hills we're looking for have something of interest. This planet is the most monotonous I've seen yet. Where it isn't jungle it's swamp; and the only difference between the two is that the jungle grows higher trees." McLaughlin's face crinkled into something like a smile, and he sat up once more.

"There's one other difference," he remarked.

"What's that?"

"In the jungle, dressed and equipped as you now are, you might live as long as a day. In the swamp, five minutes would be an optimistic estimate." Sulewayo looked down at the shorts and boots which constituted his costume, and shrugged.

"I admit the point, but I don't expect to go out this way. What I actually wear and carry, beside my professional equipment, is up to you. Also, I was referring to appearances. Beta Lyrae Nine looked almost as dull as this world from above, and I'll bet it was as least as deadly when you reached the surface." McLaughlin had never visited New Sheol, and admitted it, but it took more than that to stop Sulewayo.

"Actually, I was hoping that these hills didn't turn out to be so covered with soil that any fossils would be yards underground at the best. Do you recall any places where the rock strata themselves were exposed—steep cliffs, or deep stream gullies, perhaps?"

"Definitely yes. The big river cuts right across the range, or else starts in it. It comes out from a canyon like that of the Colorado on Earth, though a lot less spectacular. Actually I don't know anything about the country more than a couple of miles up that canyon. I was stopped on the river by rapids, and couldn't get my amphib out on either side. For the most part there simply wasn't any shore, just cliff."

"Quite a current, I suppose?" Lampert cut in.

"Actually, not very much. I went swimming in worse, on Earth."

"That hardly ties in with steep cliffs and a river cutting through a mountain range."

McLaughlin shrugged. "You're the geologist. Look it over for yourself. Maybe you'll just have to add it to the list of things you don't understand about Viridis."

"Fair enough." The pilot-commander-geophysicist nodded. "I did not mean to imply that you were not reporting accurately; but the situation you have described would be a trifle queer on more planets than Earth, I assure you. Still, with luck your cliffs will show fossils. Maybe we'll solve one problem in exchange for another. Life could be worse."

"Just hope we don't solve the first one by proving that certain geophysicists have been talking through their hats," the hitherto silent Krendall remarked.

"Eh?"

"What would you do if we found a chunk of, say, pegmatite with radioactive inclusions that checked out at half a billion years instead of the thirty-odd million you lads have been giving us as a time scale for this mudball?"

"I should check very carefully under what circumstances and in what location you found it. If necessary, I would admit that the problem had disappeared. Half a billion years would account reasonably well for the evolutionary status of this planet's life forms, though actually it took Earth a good deal longer to reach a corresponding condition. Frankly, however, I do not expect any such find. We spotted our borings rather carefully, and should have taken pretty representative samples."

"I'm sure you did. If your results are right, it just means that the problem belongs to Hans and me—and String here had better find us a lot of fossils."

"You'll have to find your own bones," McLaughlin replied. "I'm taking you to the sort of ground you want. A fossil would have to show its teeth in my

face before I'd recognize it—and then I'd probably shoot before I realized it was dead."

"All right," Sulewayo chuckled. "You take care of the quick, and Krendall and I will worry about the dead. Dr. Lampert can figure out how old the fossils are if we find any, and Take can look for stone axes."

"Or automobiles, or pieces of space-drive tubes, or other artifacts," Mitsuitei answered the implied dig. "I plan to sit back and loaf, unless and until one of you lads turns up a skull that could have held more than half an ounce of brain. I am going to be very unscientific. I believe that there is nothing on this planet for an archeologist to do, and I am not going to work myself into a lather to prove myself wrong."

"You've formed an opinion rather early in the game," Lampert remarked. "After all, remarkably little of this world has been explored. Why should there not be traces of occupation in unknown areas such as we are about to visit?"

"Because, while most of the planet remains unexplored, a very large number of places which should have furnished traces of habitation have failed to do so. We've surveyed many spots which were, or are, ideal for cities based on ocean commerce, or market centers for what could be farm areas, or spaceports. After a while you get to a point where such finds can be predicted with some certainty. As I said, I am far from certain, and it would be most unreasonable to say I was; but in the area we are seeking, I see no reason to expect anything of interest to my profession."

Lampert shrugged and brought his full attention back to the controls. The sun was slowly sinking, bringing into bolder relief the irregularities of the ground as their shadows lengthened. However, these irregularities were still few, and the jungle roof was for the most part evenly illuminated. As McLaughlin had expected, there was nothing that could be used as a landmark. In its own way, the forest was as featureless as the ocean. The pilot kept his gaze riveted ahead, in expectation of the river which the guide had told them to expect; and presently he saw it. Reflecting the color of the

faintly purplish sky, it stood out fairly well against the gray-green of the jungle, once they were close enough to penetrate the ever-present haze.

With McLaughlin nodding silent approval, Lampert swung the helicopter to the left and proceeded more nearly straight north, angling gradually toward the river. Now the jungle took on a little more feature, though still nothing that could be used for guidance. At fairly frequent intervals a glint of water became visible through the trees directly below them. Evidently numerous tributaries were feeding into the larger stream; but none of these could be seen from any distance. For the most part they were so narrow that the trees growing on each side met above them.

"I should think that one could cover a great deal of that territory in a boat," remarked Mitsuitei, after nearly half an hour in the new direction.

"You'd need an amphib," replied the guide. "A boat is all right for the main stream, but all that stuff coming in from the sides is so shallow that you'd never make progress with anything else. I've tried most of them in my own croc. Every time I've had to crawl rather than float before I was a mile from the river."

"How is the ground? Swamp?"

"No, it's fairly solid for the most part. It doesn't show very well yet even with the sun as low as it is, but the general ground level is pushing up slowly all along here. We'll be in sight of your mountains before too long."

This declaration brought all members of the group to the windows, all five pairs of eyes covering the quadrant of vision below and ahead. The meandering river was now on their left, but just visible through the haze ahead of them was the eastward turn McLaughlin had predicted. Lampert headed a little more to the right in an attempt to cut the final corner, but the helicopter reached the winding purplish band before their goal came in sight in spite of this effort. The flyer hummed on.

The bars of sunlight admitted by the side ports had been nearly horizontal when the turn to the east cut them off. They were only slightly more so when McLaughlin gave a satisfied grunt, and nodded forward. The others followed his gaze.

Straight ahead, little could be seen because of the "bright spot" familiar to every flyer—the shadowless area directly opposite the sun, centered on the aircraft's own shadow. To either side, however, the promised hills rose out of the jungle to heights exceeding the present flight altitude of the helicopter. Presumably the canyon from which the river was supposed to emerge lay in their path. So, at any rate, Lampert remarked; and McLaughlin confirmed him.

"I'd cruise pretty slowly from here on," the guide added. "There are a number of hills on this side of the range. Even if you're not worried about running into one of them, you may want to examine them for exposed rocks."

"Mightn't it be better to find a spot to park before the sun goes down?" countered the pilot.

"It might. What I said still holds, though. You haven't much chance finding one inside the canyon without quite a long search, and it will be best to stay this side of the range until sunrise. Remember my trouble in finding a beach for the amphib while I was inside."

"All right. Can we land in jungle, though?"

"Not unless you want to fold the blades in flight and drop the last twenty to fifty feet. Hunt for a fairly high hill. They're usually somewhat bare on top, and you'll at least have room for the rotors to swing. If you don't like that, or can't find a suitable hilltop, land on the river and tie up to the shore—but again, don't try that in the canyon. You're unlikely to find anything to tie up to."

"This machine has good lights, I suppose you realize—but then, you know the planet. As far as I'm concerned, what you say goes. Are the chances of a hill equally good on either side of the river?"

"Maybe a little better to the north. The ground looked higher that way when I came out of the canyon." Lampert obediently eased the flyer's course a trifle to the left, and everyone aboard watched the ground as it began to rise toward them.

At first the "hills" were merely low mounds, as jungle-covered as the level ground; but very quickly these gave way to higher, steeper rises on whose tops the larger trees grew very sparsely. One of these was quickly selected

after a brief, questioning glance from Lampert to the guide, and the helicopter began to descend.

"We'd better take what we have now." McLaughlin amplified the nod with which he had answered the pilot. "This belt of hills is pretty narrow, and we'd be into the main range in another minute or two."

"Do you know whether the other side is as abrupt, or whether—" Lampert's question was cut short by an exclamation from Mitsuitei.

"Rob! Hold it a moment!"

Lampert was a good pilot; the increase in rotor-blade pitch under his deft fingers brought the helicopter's descent to as nearly an instant halt as was possible to anything airborne. Not until he had also checked horizontal drift did he look in the direction the archaeologist was indicating. By then, everyone else had seen what had attracted Mitsuitei's attention.

Between the hill on which Lampert had intended to land and the river were several lower eminences. These were now almost directly south of the helicopter, and every detail upon them was shown in exaggerated relief by shadows stretching to the east. It was one of these hills which Mitsuitei was examining with the utmost care.

It was covered with jungle, like the rest; but a curious regularity was visible. The trees appeared, at this distance, to be of the usual species; but some of them towered over their fellows by a good thirty or forty feet.

This in itself was not odd. The whole jungle was studded with such projections. However, on this hill the taller trees seemed to have been planted in orderly rows. Five solid lines of them were visible, extending roughly north and south so that their long shadows made them stand out sharply. They were separated from each other by perhaps a quarter of a mile. Running at right angles to them were other, less outstanding rows of vegetation. Lampert was not quite sure that these were not the product of his own imagination, since the trees which formed them rose little if any above the general level. The whole hilltop, however, suggested something to every man who saw it. The archaeologist was the first to give voice to the impression.

"That was a city!"

No one answered. Some of the scientists must have thought that he was jumping from one opinion to its direct opposite on the strength of some rather feeble evidence; but the thought went unvoiced. They simply looked—except for Sulewayo, who moved to turn a camera on the scene.

"Rob! Can we land there? Now?" Lampert had anticipated this question, but could have answered it without hesitation in any case.

"Sure—if you don't mind using String's method of folding the blades and falling in." The archaeologist turned to the guide.

"Will it be hard to get there on foot from this hill we're heading for?" McLaughlin shrugged.

"From two hours to a day, depending on undergrowth."

"We have torches. We can burn our way if the vegetation is dense."

"Half a day, then. You'll still have to let the steam clear pretty often. There's little wind below the trees, and the air is saturated."

"Well, that place will be worth more than a day of anyone's time. Maybe tomorrow we can—"

"Hold up a moment, Take!" Lampert cut in, before Mitsutei could develop his plan further. "If you take String out to that hill before take-off tomorrow, what do the rest of us do for the day—or week—before you get back? What we'd better do is note this place, go on to the canyon, set up camp, get the fossil hunting going, and *after* our routine is set up and we know the more common dangers of the neighborhood, perhaps we can spare McLaughlin for a day or two so that you can look over your city—if that's what it is."

Lampert's last few words banished the hurt expression from the little man's face.

"What do you mean—*if*? What else could make a pattern like that? It must have been streets."

"Or a joint system in the rock below, trapping enough water—or draining enough off—to permit superior growth along the joint lines. Or a system of tilted strata doing the same thing—"

"If it's the latter, it's just the sort of thing you want, too. It should bring fossils near the surface."

The pilot nodded slowly. "You do make it sound more attractive. Still, I think we'd better follow the original plan, except that I may come with you myself when we do get around to looking that hill over." He turned back to the controls and resumed their descent. Mitsutei subsided once more to his seat. The archaeologist realized the wisdom of Lampert's decision, but did not particularly enjoy the enforced wait. His face showed the fact, until Sulewayo opened the camera he had been using and passed him the sheaf of prints on which the "city" appeared. As the young paleontologist had expected, these so occupied the little man's attention that he did not even notice the landing.

The helicopter settled to the hilltop which Lampert had chosen, in the center of a quadrangle of trees growing just far enough apart to give clearance to the rotors.

The sun was nearly gone. It had vanished in the haze as they dropped below flight altitude. McLaughlin knew that in all too short a time it would be as dark as Viridis ever became. The nights could be dangerous. There was quite enough light to deceive a man into thinking he could see clearly, and an inexperienced wanderer might not realize until too late that details were not really distinct and that there was no clue to direction in the shadowless glow. McLaughlin himself could use the moons, but he doubted that any other member of the party could do so. They—or their motions—took knowing.

He was pleased to note that there was no general rush to the door as the great blades whistled gently to a stop. The scientists turned to him, but remained where they were. No words were spoken, but Lampert's relinquishment of command was evident. McLaughlin unfolded his length from the seat.

"There are two choices," he said. "We can sleep in the 'copter, or outside. The first will be a trifle cramped, but the second will require either a double circle of charged wire or two armed guards on constant watch. With no offense meant, I doubt that anyone but myself in this group could qualify as a night guard."

"Why a double circle of wire?" asked Lampert.

"The wire will stop only an animal in control of its motion when it makes contact. If a Felodon were to spring from a little distance, it might not like the wire—but it could hardly stop until it reached the ground, and there should then be a similar barrier ahead of it."

"We could use a lethal voltage."

"Even if you want to take the risk—what is lethal to a Felodon will be equally so to a man—you'll have the insulation problem. There's always a darned good chance of rain before morning, and—"

"We might as well stay inside, then. We have the electric equipment, but it will take quite a while to set it up; and it hardly seems worth the trouble for a one-night stand. As you say, it will be a little crowded here. But we've all slept under worse conditions. Would anyone rather set up the fence?"

There was no answer to this question. At Lampert's direction a meal was served and eaten. Then the scientists settled down for the night, some to sleep at once, others to review plans or recheck equipment. Mitsuitei occupied himself with making careful measurements of the photographs he had been given; he was the last asleep....

Scores of miles to the southwest, the Felodon reached the river. It was no longer on the coast; some time since it had swerved inland. A casual compass check would have revealed that it was still heading straight for the now grounded helicopter. Even McLaughlin could not have told what led the creature on, familiar as he was with the animals of Viridis; but no one who had watched the thing since the flying machine had passed could have doubted its goal. Actually, it was now on the same bank of the river as the helicopter; but whatever guided it pointed across the great stream.

Without hesitating, the amphibid plunged into the water.

III

The men were awake well before sunrise. The human body takes a long, long time to accustom its physiological cycle to a change in something as

fundamental as the length of day. But they did not attempt to resume flight until the green star was once more in the sky. Mitsuitei put forth a tentative suggestion that the interval be spent in a visit to the "city" site he had seen the night before, but McLaughlin vetoed it.

"Going on foot through the jungle at night is a fool's game, though I admit people sometimes get away with it. I could get you there, but even if we turned around and came back immediately there'd be a lot of time wasted. Dr. Lampert went over all that last night. Look, that hill of yours is right by the river. After we're set up in the main camp, it will be relatively easy to drop down to it. We have collapsible boats. Unless we camp above the rapids, you won't even have to fly. Even if we're farther upstream and do have to use the 'copter, the trip will take only a few minutes."

Mitsuitei had agreed, though with evident reluctance. No one else had any desire to go out; there was not enough rock exposed on the hilltop to excite the paleontologists, the hill itself presented nothing unusual to Lampert's geophysical eye, and McLaughlin was in no hurry to get to work. They waited, therefore, until the "Claw"—Lampert had recalled Beta Librae's Arabic name—had risen and the skyglow been replaced by its emerald brilliance; then the journey was resumed.

It took, as McLaughlin had said the night before, only a few minutes. The hill where they had slept was less than five miles from the face of the mountain range. Only the haze of the night before had prevented their seeing it. The river emerged from a canyon some fifteen hundred feet in depth, a couple of miles to the south of their eastward course line.

Lampert, in hopes that the usual haze might not be too evident at this hour, climbed above the level of the cliff top to get an idea of the mountain range as a whole; but he was disappointed. For nearly an hour he cruised over the area, now several thousand feet above the western cliffs and then well below them. It slowly became evident that the range represented a single block, which had been tilted upward on the west side. The opposite slopes were very gentle, merging so gradually into the general peneplain level of the continent that it was impossible to say decisively just where the range ended. The river did originate somewhere beyond the range, cutting entirely through it, and as the guide had said, its current was not particularly swift. Lampert had much explaining to do. After all, water should have drained toward the low side of the block.

"It seems evident," he summed up his ideas as they hovered once more over the western cliffs, "that the river was here before this particular bit of block tilting occurred. This planet does have some diastrophic forces left in its crust, in spite of its generally smooth nature. Apparently this just represents the end of a long period of rest, such as the earth has had several times. As a matter of fact, I have no business calling it the end of such a period; it might be fifty million years before the world will be generally mountainous again."

"Why do you say *again*, Rob?" asked Krendall. "According to findings of your own colleagues, this planet has hardly been solid for forty million years. Could it be this flat now if it had ever been markedly mountainous in that time?"

"Good point. I don't know, but would be inclined to doubt it. Well, we'll cancel the 'again' if it will make you happy. In any case the block forming this range came up slowly enough so that even this river, with its relatively low cutting power, was able to keep pace with it and not be deflected. Probably—" he glanced at Mitsuitei—"the rock of which it is made will turn out to be quite strongly jointed. It looks rather that way from above—the river course, I mean. A lot of right angle, or what were once right angle, bends."

"We'd better go down and look for a camp along the river somewhere," put in Mitsuitei. "Let's start at the cliff end. Then we may wind up reasonably close to that hill—and I still want to look it over, joints or no joints."

"Fair enough." Lampert eased the helicopter once more downward until they were only a few hundred feet above the jungle, moved along the cliff face until they reached the canyon, and, very cautiously, entered. His caution proved unnecessary. The air currents in no way resembled the treacherous hodge-podge he had expected, at least not over the center of the river. A steady wind was blowing into the canyon mouth, but did not seem to be eddying very much even at the numerous bends.

To the archeologist's annoyance, two sets of rapids were passed before a place was reached where the bank was wide enough for a camp site. At this point a fairly large side canyon entered the main one from the north. Where

its central stream joined the main river a gravelly area several acres in extent offered itself for the purposes of the scientists. Lampert brought the helicopter down on this surface. The surroundings looked promising; the cliffs facing both canyons looked reasonably accessible on foot for some distance, at least along their bases. Climbing appeared to be impracticable for the most part, as the rock walls rose sheer except for the occasional joints which Lampert had predicted; but the material was certainly sedimentary, and everyone but the guide tumbled out of the flyer with a glow in his eyes which promised a speedy scattering of the party.

With some difficulty, McLaughlin got them together. A site, some twenty yards square, was selected against one of the cliffs and fenced off. The big, prefabricated sheet-metal "tent" was erected and its tiny conditioning unit installed; sleeping and cooking gear were placed inside. That completed, geologist's hammers appeared as though by magic; and McLaughlin realized that he had better do some explaining before he lost a scientist or two. Once more he called them together.

"All right, gentlemen. I admit the necessary camp work has been done, and there should be nothing to keep you from your projects. Still, there are some things you had better understand.

"Having canyon walls on all sides does not make this place safe. Every carnivore and poison lizard on this planet could get to us by way of the river—even the ones which look like land animals. Every one of them could swim under water from a point out of sight in either direction to where you are standing; and if you think he would have to come up at least once to judge your position, guess again. I don't know how they do it, and neither does anyone else; but a Felodon could submerge around the bend up there, come up behind the helicopter out of sight of any one of us and be waiting when we marched around the machine. Therefore, go armed at all times. I know you want to cover a lot of ground, and can't stick in one party; but I insist that you do not go anywhere alone. Take at least one companion. Preferably one who is not a member of your own field. If you two paleontologists are together, for example, it seems more than likely that you'll be found with your heads in the same hole in the rock. When one of you has to dig, make sure the other has his neck on a swivel. I know this will slow your work, but not as much as if the work had to wait for a new investigating team from Emeraude—or from Earth.

"You've seen most of the dangerous animals in the zoo at Emeraude, so I won't waste time describing them. Just remember that you won't always hear them coming. You'll have to use your eyes.

"All right, Dr. Lampert. You're the boss, as far as the scientific work goes. Who does what, and where?"

The geophysicist gave no sign of having detected the humor in the guide's remark, but began speaking at once.

"I should say that the main canyon upstream and the side one in the same direction should be covered first. We've already used up a good deal of today, and would waste more breaking out the boats. Ndomi and I will go up the main stream; Hans and Take can take the other. Don't hurry. If anything looks good, take the time to investigate it on the spot. Of course, if it is obviously a major job, just mark it and go on. There's no sense in one man's trying to exhume a six-foot lizard skull.

"Since this region must have been sea when the limestone was deposited, there's not much chance of land animals. However, we want as complete a chronological series as possible, so do the best you can on this level. We'll try for higher formations later. There should be plenty farther upriver, if this block is tilted the way it seems to be.

"String, perhaps you'd better go with Take and Hans. Set out when you're ready. Be back in—" he glanced automatically at the narrow strip of purplish-blue sky, then at his watch—"four hours; then we'll compare notes. After that we can either concentrate on one place or the other, or break out the boats and cross the streams, as indicated."

Twenty minutes later the parties were out of sight of each other and the helicopter. Lampert had spent the first few minutes of the walk wondering whether he had been too obvious in arranging for both the guide and Krendall to accompany the little archaeologist; but he quickly convinced himself that McLaughlin's speech had covered the arrangements pretty well.

In any case, he would probably have been distracted soon enough. The cliffs were interesting. Limestone, evidently, as expected—but rather dense, at

that; maybe some barium replacing the calcium? or was the gravity different enough to destroy his judgement for such a small fragment? Probably not. He was actually using inertia more than weight in making his estimate. Anyway, the stuff was certainly a carbonate. It frothed satisfyingly under a drop of acid from Lampert's kit.

And there were fossils. Sulewayo's form was bent over a spot on the cliff face, examining minutely; but Lampert could see others from where he stood. None seemed remarkable. Most were rather evidently shellfish. He carefully refrained from giving them names according to the genera they resembled in Earth's rocks; Sulewayo and his colleagues frowned on the practice, which could be most misleading. He could not, however, resist the temptation to think of them as scallops.

"What do you have there, Ndomi?" He knew the other would not have spent so long on any shellfish.

"Not sure, precisely. Maybe vertebrate, maybe not. What could be armor and what could be ribs all mixed up. I think I'll mark it for future reference."

"I suppose it'll be another Devonian whatsit, like everything else on this planet, when you do decide."

"Pennsylvanian would better describe the world as a whole. Barring that, you may be right. Rob, if you'd give me a hand here we could get some basic work done."

"Eh?"

"You say this is a tilted block. In lowest formations right now. I'd like to get photos and if possible specimens of as many different varieties of shellfish as possible, at each level. Then it may be possible to set up some sort of temporal sequence—and use the things as index fossils if animals do evolve on this be-nighted mudball. If you could get me some radioactive dates at two or three nicely scattered levels, it would also help."

"Thanks," returned Lampert drily. "I could use material like that myself. I can tell you what you probably already know—you're not likely to get anything of the sort from limestone."

"Well—intrusions are always possible."

"You watch for 'em, then." The pair went to work.

Two hours out, a little more than one back. There was no one at the helicopter when they reached it, but the other group came in only a few minutes over the four-hour limit which Lampert had imposed. A comparison of notes over the meal which had been quickly prepared indicated that the second group had gone farther in point of miles covered, but had accomplished less work. Krendall had had the same idea as Sulewayo. But he had not attempted to carry it out since his canyon did not cut across the range, and would presumably not furnish a continuous change in formations.

Lampert and Sulewayo, as it happened, had not found any evidence of change themselves. The last fossils they had found were at least superficially identical with the first. There was the usual evidence of bedding, and it had been quite evident geometrically that the walk had taken them to originally higher, and presumably later, levels; but in what must have been eight hundred feet or more of original deposit, there seemed to have been no significant change in the fossil life. What eight hundred feet would mean in point of time, of course, no one had the least idea. There was not even a good guess as to how fast carbonates might be expected to precipitate in a Viridian ocean. Anyone could compute the carbonate ion equilibrium between atmosphere and sea, but no one knew anything to speak of about carbonate-precipitating organisms of the planet.

Mitsuiei changed the subject slightly at this point.

"We found several of the joints you predicted," he said to Lampert.

"Oh? Very wide? We didn't spot anything that was obviously a joint. But there were several small side canyons—all narrow enough for us to wade or jump their central streams—which might have started life that way."

"Ours were quite narrow, and bore traces of volcanic ash at the bottoms."

"Eh?"

"That's right, Rob. Here's a bit of it I brought back. I thought you might want a little corroboration on that one." Krendall handed over a bit of crumbly tuff as he spoke. Lampert examined it with pursed lips.

"Maybe we'd better get back into the air, and search the neighborhood for volcanoes," he said at last. "I can't bring myself to believe in two full

mountain-building cycles on this planet—and if I could, I'd have a hard time swallowing the idea of these limestone layers coming up, going down, and coming up again unaltered. How deep were these volcanic deposits?"

"Variable. Shallowest in the wider joints; in the very narrow ones, up out of sight."

"Suggesting that they've been washing out for some time since the original settling. Anything organic in them?"

"Nothing turned up yet."

"Do they extend below the present river level, or what?"

"They're at least down to it. We couldn't do any major excavating."

"If they run much below," muttered Lampert, "I'll join the roster of geophysicists who have been driven off the rails by this woozy world. Well, let's assume as a working hypothesis that the volcanic activity is relatively recent. That will at least have the advantage of keeping me sane, until something comes up to disprove it." He finished his meal in silence, while McLaughlin gave a reproofing lecture on the matter of wading.

There was still a little daylight to go when all the men had eaten; and Lampert, Sulewayo and the archaeologist took the helicopter up the main canyon to check on the possibility of walking to any really new deposits.

They were sure, from changes of color already seen at various levels up the cliff face, that these existed. But it appeared that the lowest of them did not reach river level for more than a dozen miles. The distance was less mapwise, but the canyon, winding back and forth around what the geophysicist still felt must be joint-bounded blocks, went a good two miles in other directions for each one that it led eastward. Realizing this, the explorers lifted the helicopter and began checking as close to the cliffs as Lampert dared at higher levels. In this way they worked back toward the camp site. Once again it was Mitsuitei who first spotted something of major interest.

"Found another city, Take?" asked Sulewayo at the other's call.

"Not exactly. It's—well, I guess it's really a system of those joints you keep talking about. Still, it looks awfully regular." He sounded a little wistful.

"It does." The paleontologist nodded slowly. "As you say, it's probably a joint system. Also, it's probably full of volcanic ash, if my eyes don't deceive me. Rob, what's the chance of a landing on one of the shelves? There are at least three formations accessible on foot from that point; and I could get some more tuff samples to make or break your peace of mind, while I was doing my own work."

Lampert examined the area carefully. Like Earth's Grand Canyon, this one receded from time to time in shelves where softer layers of rock had worn further back, or the orogenic processes had paused to give the river a longer bite at that level. The cracks Mitsuitei had seen formed a neat crisscross pattern on the top of one of the shelves. Some of them betrayed their nature by emerging from its vertical face. It was admittedly an unusually small-scale joint pattern, at least for this mountain system, and might well contain readable evidence of the forces which had shaped the area.

However, they had only one helicopter. Lampert slowly shook his head in negation.

"I'm afraid not, Ndomi. Your shelves may be big enough, but they're not level enough. I'd have to make a swinging landing, and I'm not that good a pilot."

"Well, how about letting me down on the ladder? We have a hundred feet of that, so you could be up above the next shelf while I went down. You'd have plenty of blade clearance. That next level goes back a couple of hundred feet."

"That might be all right." Lampert spoke hesitantly. "You certainly have the right to risk your own neck on the climb if you want to. We won't try it tonight, though. I'd like to check with String on the advisability of your being there alone. The place looks pretty hard to reach for anything that doesn't fly, and I don't know of any really dangerous flying things on this world; but we'd still better check."

"All right with me. I'd just as soon have a full day, anyway."

"If Ndomi will be spending a day alone up here, how about having String take me to the other place, and settle that point once and for all?" asked

Mitsuitei as the helicopter eased downward toward the camp. "That would still leave Hans and you to form another team for whatever else you want to do."

"That should be all right. It'll depend, though, on whether String thinks it's safe for a man to work alone on that shelf."

The proposition was put to McLaughlin as soon as the machine was landed. To Lampert's surprise, the guide gave a qualified approval.

"Remember," he concluded, "I don't know what lives on the cliffs. It's country I've never covered. All I'm saying is that no Viridian animal I know of could get there, except flying ones; and they're nothing to worry about, especially in the day-time. I'd like to go with you to look over the place when you take him up tomorrow, and strongly recommend that he carry a communicator as well as a weapon; but unless I see something you haven't mentioned when I do go, I would say it was all right...."

Once more the Felodon reached the river, but this time it did not cross. It was no longer heading straight for the helicopter. Hills had not altered its course, but the cliffs had. They formed a wall on its right which was too nearly vertical for its agility and strength. Even this barrier, however, had caused no visible hesitation or doubt. It had swerved, followed the base of the wall to the point where the river emerged and plunged in as promptly as it had done before. Few amphibians have ever lost the art of swimming when their larval gills vanished; the feeble current meant nothing to the Felodon.

It turned upstream and went on its way.

IV

Ndomi Sulewayo had pursued his occupation on terraces of Earth's Grand Canyon, on cliffsides of Fomalhaut Four's highest range and in badlands on the dimly lighted Antares Twelve. The physical hazards of his present position troubled him little. McLaughlin had agreed that the ledge where the paleontologist had been left was inaccessible to the larger carnivores, and had merely issued a final warning about poisonous "lizards." The primary danger, as nearly as Sulewayo could see, was that something might happen

to the helicopter. He certainly could not rejoin the others on foot. He was facing a sheer wall some sixty feet high. A score of yards behind him the terrace ended in another straight drop of several hundred feet. A quarter of a mile on either side, the flat surface ended; to the west, by narrowing until the two walls became one; at the other end, it was cut off as far as he was concerned by a joint penetrating apparently the full depth of the canyon.

There were several other cracks in the wall facing him. Like those in the tributary canyon explored by Krendall and Mitsuitei, these were packed with volcanic detritus. This was hard to reconcile with the suggestion that erosion had been long at work. In such a case, the higher portions should have washed away long before the material found at the canyon bottom.

Examination at close range suggested a possible explanation. The tuff at this point was fairly well cemented. It seemed reasonable to suppose that the joints had been present before the mountains had started to rise; that a volcanic mud flow had filled them with detritus; that the new material had then been cemented by dissolved material coming from above. This would make the top levels of the tuff more resistant than those lower down, where the cementing minerals had not reached, and account for what had been seen so far.

The hypothesis also implied a plentiful supply of fossils. Volcanic mud flowing into a crack in the ground should carry plenty with it. Sulewayo set to work with a hammer, and was presently soaking with perspiration.

He was tempted to remove some of his clothing; but this had been treated chemically to repel Viridian insects and caution prevailed. McLaughlin had not mentioned any dangerous biters or stingers, and in all probability his blood would not be to the taste of any such creatures on this world—but if the mosquito or tick did not learn that fact until after it had tried, Sulewayo would hardly profit by it. In any case the temptation to strip passed quickly. In only a few minutes, his attention was fully occupied by his work; for the expected fossils proved to be present in very satisfactory numbers.

Most seemed rather fragmentary. Apparently the original creatures had been tumbled about rather badly before the medium hardened. However, the remains were definitely bones, as he had expected and hoped. For some time Sulewayo was occupied alternately digging out more fragments and trying to fit the more hopeful-looking specimens together, although he had no success

at the latter job. Then evidence of a more complete set of remains appeared, and he instantly slowed down to the incredibly meticulous procedure which marks a paleontologist anywhere in the universe.

At this time he had cut perhaps a foot into the tuff for the full three-foot width of the crack and from terrace level up to about his own height. In spite of its apparently firm texture, the rock was extremely soft; and the old question about erosion was reappearing. Big pockets of extremely crumbly material had been responsible for most of his speed. Now, however, with the usual perversity of the inanimate, a firmer substance was encountered, apparently encasing the bones he suspected of existing a little farther on. This combined with his increased care to bring almost to a halt the removal of rock from the cleft.

The bones were there. Perhaps they had been betrayed by a discoloration of the rock too faint for him to have noticed consciously; perhaps something more subtle is involved in the makeup of a successful field worker in paleontology, but as flake after flake of the matrix fell away under his attack a shape gradually took form.

At first a single bone which might have been an unusually short digit or an unusually long carpal—or, of course, something totally unrelated to either—was outlined. Then another, close enough to suggest that their lifetime relationship might have been maintained. And another—Sulewayo failed to hear the approach of the helicopter until its rotor wash from a hundred feet above lifted the dust about his ankles.

Knowing that Lampert would be having trouble holding that close to the cliffside, the paleontologist reluctantly hooked his equipment to his belt and started up the ladder. Five minutes later they were back in the camp, with Krendall listening eagerly to Sulewayo's description of his find.

"It's certainly a vertebrate, Hans. That stuff can't possibly be shell or wood. It's almost certainly a land dweller—"

"Likely enough in that sort of rock, anyway."

"—because I got enough uncovered to be nearly certain that it's a foot. Certainly a limb that would not be needed by a swimmer."

"Like an ichthyosaur?" queried Lampert innocently. Sulewayo grinned.

"Quite possibly. More likely one of our ubiquitous amphibids, though. Certainly something worth getting out, since the general idea is to get an evolutionary sequence of some sort."

"I suppose that means you'll want me to date the eruption which filled all these cracks with detritus, then."

"Sure. But there's no hurry. Tomorrow will do." Lampert found he had no answer to this, and Mitsuitei managed to edge into the discussion. He had spent the day with McLaughlin, as he had hoped; and mere failure to find paving stones had not damped his ardor.

"I suppose you and Hans will both want to go up the cliff tomorrow," he remarked. "In that case, Rob might as well stay with String and me. It will speed up the digging back at my hill."

"Are you still scraping dirt off that thing?" asked Sulewayo in mock surprise. "Didn't one day indicate that it was a joint pattern like the rest?"

"Not yet. We haven't gotten down to rock over any place where your cracks should be. The root tangle of the taller trees slows the digging. I admit the rock is limestone like the cliff, but there's still no evidence why those trees grow so regularly."

"That's just what we've been saying all along; but you keep looking for the remains of a city."

"I gathered, Ndomi, from your recent conversation that you were digging for a land animal on the basis of three bones. Either you are working on hunch, which destroys your right to criticize, or you are reasoning from knowledge not available to the rest of us. In the latter case, you should be at least open-minded enough to credit me with equivalent knowledge in my own field."

It was Sulewayo's turn to have nothing to say; he had honestly supposed that the archaeologist had been taking the "city" hypothesis no more seriously than the rest. He apologized at once, and peace was restored. Lampert sealed it by agreeing to Mitsuitei's suggestion.

The rest of the evening was spent in detail planning by the two groups. At sunset, all turned in to sleep behind the protection of the electrified fence. Even the guide regarded this as an adequate safeguard.

Apparently his opinion was shared by at least one other. The Felodon had spent most of the day under water, part of the time in the canyon fairly close to Lampert and Krendall and later down the stream by the site where the guide and archaeologist had been working. At neither place had it emerged, or shown the slightest sign of wanting to attack. McLaughlin's reference to the strange instinct of the creatures seemed justified. It certainly could not see the men, but just as certainly was aware of their presence.

What it was about the alien visitors which exercised such an influence on the minute brain of the carnivore, no one could have said—then. Any watcher who had supposed, from its earlier actions, that it was moved by a desire for new and different taste sensations would have had to discard the notion now.

With the men safely settled down behind their fence, the beast suddenly turned back downstream. It had returned to the camp site at the end of the working day. In an hour it was in the jungle below the canyon; in another it had killed, and was feeding as it had the moment before the hum of the helicopter had first attracted its attention. This time it finished the meal in peace; and once finished, did not show immediate signs of its former obsession.

Instead it sought a lair and relaxed, blending so perfectly into the undergrowth and remaining so silent that within a few minutes small animals were passing only feet away from the concealed killer.

Robin Lampert was only a fair statistician, but if he had been acquainted with the moves of that Felodon during the last few days, even he would have been willing to take oath that more than chance was involved. He would probably have wanted to dissect the animal in search of whatever mechanism was controlling it.

But Robin Lampert knew nothing of the creature. Neither did Takehiko Mitsuitei; and that was rather unfortunate, for the lair it had selected was on

the same hill as the archaeologist's digging site, and a scant quarter mile away from the pit Mitsuitei had left.

The rising of the green sun was not visible the next morning. The ever-present mist had thickened into a solid layer of cloud, and hissing rain cut the visibility to a few hundred yards. The helicopter felt its way down to the hill with radar, landed on the river, taxied on its floats to the bank and was moored. Lampert, McLaughlin and Mitsuitei emerged, the scientists laden with apparatus, and started up the hill toward the site. The guide carried only his weapons.

The equipment was not of the sort Mitsuitei was accustomed to using. It actually belonged to Lampert. Normally it would not be used in an archaeological dig, any more than it would have been had they been fossil hunting; for neither activity takes kindly to any sort of automatic digging machinery. Lampert had suggested its use, however, in order to get a rapid idea of the nature of the soil cover, bed rock and joint structure of the hill. If evidence warranted, it would be abandoned for the slower methods of digging. If not, a few hours would permit them to learn as much about the area as many days of work with slower equipment.

The hole Mitsuitei had already dug was part way up the hill, in a space cleared of underbrush by a flamethrower. Several other such clearings were in the neighborhood. As the archaeologist had said, he had made more than one attempt at digging which had been frustrated by roots.

Somewhat to Lampert's surprise, it was possible to tell even from ground level the orientation of the taller trees which had been so prominent from the air. Even the smaller plants showed signs of some underground influence. Between the tallest trees, tracing out the straight lines the men had seen from above, the underbrush formed an almost impenetrable wall. Elsewhere foot travel was easy, though the surface was by no means barren. Lampert understood how there might indeed have been difficulty in digging on one of the fertile lines, and admitted as much.

"That's the trouble," responded Mitsuitei. "I'd like to get down right at such a point, to see what's underneath. It seems to me that paving might be

responsible, if they'd used the right materials. Lots of civilizations have used organic substances which decay to good fertilizer. Then there might be the remains of a sewage system, which would account for richer soil—"

"After the time which must have passed since the place was buried?"

"It has happened. In such a case, of course, trace elements rather than nitrates or phosphates are responsible. That's what I suspect here."

"But wouldn't it be better to dig where you actually have—in the middle of a block, if that's what it is? Then you'd be fairly certain to hit a building, which should be richer ground than a street."

"Only if you actually strike artifacts. The building itself might be much less well preserved than a paved street. However, you are the one who's handling that mechanical mole. Dig where you want, and see what you can learn about this hilltop. Just get me at least a couple of cores from my 'streets' before you're done, please."

Lampert nodded and proceeded to assemble his equipment. The "mole" was a cylinder about five centimeters in diameter and three times as long. A cutter-lined mouth occupied one end, while the other was attached to a snaky appendage which was wound on a fair sized drum. A set of control knobs and indicators were mounted near the center of the drum.

The geophysicist set the cylinder on the ground mouth downward, pushing it into the soft earth far enough to assure its remaining upright. Then he turned to his controls and after a moment, with very little noise, the cylinder began to sink into the ground. In a few seconds it was out of sight, trailing its snaky neck after it.

The men watched it in silence. Perhaps thirty seconds after it disappeared, there was a minor convulsion in the neck, a momentarily rising hum from the machinery, and a plug of dirt about two centimeters in diameter and five long was ejected from a port in the center of the drum. This was seized by Lampert and examined briefly, then tossed aside. "The soil is pretty deep," he remarked.

"How far down did that come from?" asked Mitsuitei.

"One meter. That's the sampling interval I've set in it, for now. If it meets anything much harder or easier to penetrate, it will warn me and I'll grab them more frequently." Conversation lapsed while two more samples arrived

and were inspected. Then a light flickered on the panel, and Lampert reset one of his knobs; and almost immediately a core of light gray limestone was produced.



"Apparently the same stuff as the cliffs," said Lampert after examining the specimen. "Do you want to go any deeper, or drill a few more holes to get an idea of the contour?"

"How fast will that thing go through limestone?"

"A couple of centimeters per minute. It's too small to pack a real power unit."

"Give it five minutes, just to make sure it isn't a building block."

"Ten centimeters wouldn't give you a whole building block."

"A sample from that far inside one would tell me what I want to know. You rock-chippers don't seem to think that archaeology is a science yet. Let me have that first core, too." Mitsuitei looked confident to the point of being cocky, and Lampert let the mole burrow on. The second core came in due time, and the little man set merrily to work with tiny chips from the two stone cylinders, a pinch of the lowest soil sample which had been acquired, a small comparison microscope and a kit full of tiny reagent bottles. Lampert used the time the tests consumed in reversing the mole and resetting the equipment on a new spot. By the time the little mechanism had gnawed its way once more to rock, Mitsuitei was forced to admit that the formation appeared to be natural.

He did not seem as disheartened by the discovery as might have been expected. He simply waited for more cores, his narrow face reflecting nothing but the utter absorption Lampert knew he experienced whenever a problem arose in his line. In spite of his apparent tendency to jump to conclusions, Takehiko Mitsuitei was an experienced and respected member of his profession. Lampert knew enough about his record to be perfectly willing to accept his instructions for the present.

A series of holes was drilled, from the original position toward one of the "streets" forty yards away from it. After each the archaeologist admitted with perfect cheerfulness that there was nothing inconsistent with the idea that the hill was a perfectly natural formation. He still insisted, however, that the regular lines of trees, reinforced as they were by the undergrowth pattern, required explanation.

Lampert admitted this, but felt that he knew what the explanation would be. After all, volcanic residue is more than likely to contain the trace elements vegetation requires, even on Viridis.

Finally the time came to get verification—or the opposite. The flamethrower had to be used this time, and for several minutes clouds of steam swirled about the men as its blue-white tongue fought the sappy, rain-soaked undergrowth. Then the mole and its controls were wheeled into place, and the little robot once more nosed its way out of sight.

"I don't suppose you want any samples above the regular rock level, do you?" asked Lampert as the machine disappeared.

"I think it would be best if we took them as usual," was the reply. "For one thing, we should try to learn the depth at which the soil composition changes—we are at least agreed that it changes in some manner, after all."

"True enough." The geophysicist set his controls, and the process continued—a process familiar now to McLaughlin as well as the scientists, for the guide had caught numerous glimpses of what was going on while he prowled about the work area on self-imposed guard duty.



Mitsuiei took the crumbly soil cores as they came, examined them quickly—they were arriving every few seconds—and filed them in numbered compartments in a specimen case he had opened. Detailed stratigraphy would come later. For some time there was no gross evidence of change in the soil; not, in fact, until his first case had been filled.

"Can you stop that thing for a moment, Rob?" he asked at this point. "I don't want to lose track of these, and will have to hold up while I open a new case."

"All right. I thought you'd want to stop for thought soon anyway."

"Why?"

"Because the mole is nearly four meters down, well below the depth at which we hit bedrock before, and is still in soil."

"Eh? But—but it's still ordinary soil; none of your volcanic ash."

"Tuff had been eroded out of a lot of the joints in the cliffs. There's no reason to expect it to be at the same level as the surrounding rock."

"That's true." Mitsuitei paused in thought for a moment. "If we keep on going straight down, we may just be working into a natural crack, as you say. Might it not be better to drill several holes within a few square yards here, to determine whether it is a narrow joint such as you expect or an actual edge to the rock at this level?"

"Maybe the edge of a roof, eh?" Lampert chuckled, but spoke in a manner which could give no offense. "I can do better than that. Don't need to pull up and start over; simply drill horizontally from where we are now. Shouldn't take long to get dimensions, if that's all you want." He halted the robot momentarily, and from a compartment in the drum removed something like a small theodolite mounting. This he set up on a short tripod over the point where the neck of the mole emerged from the ground, and set a pointer at right angles to the line of tall trees. Then he started the digging again.

V

Four starts in as many different directions and twenty minutes of time showed fairly conclusively that the line of vegetation which had given rise to the "street" theory was growing along a straight crack, apparently a fairly ordinary joint, in the limestone. While several more holes would have to be drilled to prove it, even Mitsuitei was willing to admit that in all probability the remaining lines would be found to be over similar cracks.

"You must admit, though, that the regularity of this joint pattern is pretty unusual," the archaeologist said at length.

"It's far from being unknown," Lampert replied. "I got my first large taste of it in my student days back on Earth. Fly over the mesa country in southwestern North America sometime. Most of the joints there are invisible from a distance, of course; but at the edge of a butte where weathering is most prominent the blocks have frequently started to separate, and the thing looks as though it had been put together from outsize bricks."

"Hmph. Seem to remember something of the sort myself, now that you mention it. I did some digging in that area, too. I shouldn't have connected that sort of country with what we have here, though."

"Different meat; same skeleton," replied Lampert.

"But how about this volcanic ash, or mud, or whatever it is, which at least fills the joints we saw in the cliff? That's not so usual, is it?"

"Not in my experience. But granting the joints and the volcanoes, there's nothing really surprising about it. Incidentally, we don't know that this crack we're standing on has the same filling. We'd better bore down again to make sure. At least we may get some idea of the date of the volcanic action compared to that of the orogeny that tilted the block where we're camped. If there's tuff down here too, it will substantiate the idea that the vulcanism is the older."

"Why? Couldn't ash have settled down here as well as up there at substantially the same time?"

"It could. But I'd bet a fairly respectable sum that the tuff we saw in the canyon was from a mud flow, not a fall of airborne ash. That could hardly have reached the top of the cliffs—actually, the opposite slope of the mountains, where Sulewayo is working—and this area simultaneously."

"Maybe from different eruptions? I get the impression that this world has a slight tendency to produce volcanic fields rather than individual cones or flows."

"Might be. Chemistry will probably settle that question." During the latter part of this discussion Lampert had directed the mole once more downwards, and every half meter of travel another core was added to the collection. At six and a half meters below the soil the first solid specimen arrived; the others had been held together only by roots. This one, however, caused the two scientists to look at each other. Lampert nodded slowly, with a smile. Mitsuitei gave a shrug, and let an expression of resignation play over his usually impassive features.

The core was tuff, apparently identical with that in the cliffs to the east. It even contained fossils.

"I guess this whole dig might as well be taken over by the paleontology department," Lampert commented finally. "I suppose they'll at least want to compare fossils in the tilted and level strata."

"I suppose so." Mitsuitei was turning the little cylinder over and over in his hand. "Tell me, Rob, what's this little speck of green?"

"Copper salts of one sort or another, I suppose." Lampert was not greatly interested. "A lot of secondary minerals form in and under volcanic detritus. On this world, carbonates like malachite should form quite readily."

"Why should it form in a regular thread like this?"

"You mean a vein? Hard to tell precisely. Varying rates of water seepage, varying degrees of oxygen or carbon dioxide penetration, varying degrees of compactness in the rock where the stuff is formed—"

"I don't mean a vein. This is in a cylindrical body going right through the core from one side to the other, as though there had been a copper wire there originally which had been attacked by soil acids."

"Let's see. You're right. It's hardly an ordinary vein, though your suggestion seems a trifle far fetched. The paleontologists can probably furnish an idea. Maybe a vine or even a worm buried in the mud flow acted as the precipitating agent for copper salts in the subsequent seepage—I've seen beautiful fossils of pyrite which had been formed that way."

"But this shows no trace of structure, except for its exterior shape."

"Isn't a really well preserved structure the exception rather than the rule in fossils?"

"I suppose so. Still, I'd like to know just how far, and which way, this green thread goes. I'd also like to know whether there are dilute copper deposits spread through this rock, which could be concentrated in the way you suggest."

"The first could be learned by taking enough cores. The other would call for some very careful analysis of samples which had been selected with a very sedulous eye kept on the stratigraphy. You know that; you must have done that sort of thing looking for carbon-fourteen samples, at times."

"Yes, I see that. Could you make such analyses here?"

"No, except for the mere presence of copper. The cores would have to go back to a well equipped lab. Still, if you want to get them, it's all right with me. Problems were made to be solved. I'll admit this one doesn't seem very exciting to me, but I can use your data after you finish for work of my own. You should wind up with material for a pretty complete geochemical picture of this neighborhood. Shall I get the cores for you?"

"Yes, please."

"Silly question. All right." The mole was drawn up a short distance, and sent questing downward once more at an angle to the original shaft, branching off a short distance above the level from which the copper deposit had come. Again and again the process was repeated, each time at a slightly different bearing from the central hole; and Mitsuitei examined each core for traces of green. At last he found it, piercing the little cylinder of rock as the other had done; and then, at his suggestion, Lampert reset the mole to get a sample in the opposite direction from the one which had furnished the new specimen.

This also checked positive; and four more samples, taken along the same line at various distances, all did the same.

Apparently the line of green extended for some distance, about parallel both to the surface of the ground and the trend of the joint in which it was buried. Mitsuitei was radiant.

"I'm going down to that level if I have to come back with an expedition of my own! If that's a fossil worm, it's worth getting the whole length anyway—but I don't believe it is. I—"

"That will take a lot of time, you know," Lampert pointed out mildly.

"Certainly I know! Even if I use your fast excavator down to the tuff level, I'll have to do detail work from then on. What of it?"

"Well, the others may have jobs they want to do—"

"Then they can do them! What are we here for, anyway? I thought it was to investigate the past of this planet! Ndomi and Hans are doing that their own way right now. Why can't I? I'm an archaeologist, and I came along to do

any archaeological work that presented itself to do; this is the only thing of the sort anyone's seen so far. I know what you're thinking. Maybe you're partly right. I certainly won't bet any money that this thread of green is a fossil telephone wire; but it's as likely to be that as anything else you've suggested, and I'm going down to that level and sift the whole volume. Hans and Ndomi can have any fossils I find if that will make you happier—and if one of them says he has no use for fossils he didn't dig himself, I'll make him eat his words. I can identify, locate and report on anything that turns up in a rock as well as any of those jigsaw-puzzle people; and I can do it in mud, too, which is more than any of them could manage."

"Don't get hot under the collar. If you can help it on this planet. You sound as though one of the boys had been giving you a lecture on the importance of knowing what strata a given series of specimens represent."

"Not one of our boys—they have a little more sense. But there was a young paleontologist when I was covering the Antares worlds whose memory still makes my blood pressure go up. Never mind me; that's not important. But I want to make this dig."

"It will tie up machines, however freely we can spare time," Lampert said slowly. "I'll tell you: how about this? We spend the rest of the day getting cores from other points along these cracks. For one thing, we ought to know more about the structure of the hill, and for another, we might find more of your 'wires.' After all, the chance of our hitting the only one around is pretty remote. I can't quite see a single dropped piece of copper wire showing up in the first two days of a project like this."

"I neither said nor implied that this should be the only piece. I don't doubt for a moment that there are others, whether they are wires or worms."

"Sorry. Well, we take these cores back to camp this evening, together with any others we find of the same sort, and let Hans and Ndomi look them over. If they don't turn out to be something that the boys recognize and can classify right off the bat, we come back tomorrow with all the digging machinery you want, and dig until you either find all you want, satisfy yourself that there's nothing here or find something which obviously requires more specialized attention than we can give it. All right?"

"Nothing could be fairer. Let's go!"

The discussion in camp that evening was animated beyond anything the guide had heard. His original estimate of these men as relatively quiet specimens underwent a sharp revision. Mitsuitei's report of the day's activity at his site had, it is true, been delivered quite calmly; but from then on matters grew progressively livelier. This was not caused by opposition to the archaeologist's plans. The others were all in favor of remaining, for their own reasons. However, the question of just what was likely to be found gave rise to much rather barbed comment on Sulewayo's part. "I don't see how you can expect to find any trace of civilized work here," he said flatly at one point. "The animal and plant life of this planet is at a stage of evolution corresponding to something like Earth's Pennsylvanian age, when the amphibians were the highest known forms of life. I'm not saying that there couldn't be such a thing as an intelligent amphibian. But I do say that the normal set of evolutionary forces which, on both Earth and Viridis, produced creatures of the amphibian pattern could have done that *or* produced an intelligent fish; not both. If the latter ever evolved, it failed; for the amphibians—pardon me, amphibids—are here. To get an intelligent amphibid on this world will—or would, if the sun were to last long enough—require another orogenic period with the accompanying climatic changes. Then you'd stand a considerably higher chance of getting reptiles instead, if the comparative work done on over four hundred planets carries any meaning."

"I don't doubt the value of the work at all. You are very probably correct. It did not occur to me to expect remains of intelligent amphibians. I saw no reason to pre-suppose that anything in the way of artifacts which I might find would necessarily be native to this planet."

"You think there were other visitors from outside the Beta Librae system?"

"The possibility certainly exists. Here we are."

"But for Pete's sake! Do you really expect that they stayed long enough to build a city, or do you think you have the remains of a camp like ours, or what?"

"I don't think anything. It has been suggested that such people did come, and stayed long enough to—"

"And you think you've found them."

"I think nothing, except that I have found, with Rob's help, something which neither his professional knowledge, nor mine, nor even yours, is able to explain; and I think an explanation is desirable. I hope you won't consider me discourteous for pointing out that each time you have tried to accuse me of jumping to conclusions, you have been able to do so only by jumping to some yourself. I might further add that the suggestion that this planet had been stocked with its present supply of life types by visitors from space was advanced by a paleontologist, not by one of my colleagues. I gather he could not understand how life could evolve to the state it shows in the thirty-odd million years that the planet seems to have been solid. I neither support nor deride the idea; I simply want to gather data, in an attempt to explain a much simpler question—why are narrow threads of copper compounds to be found every few feet in the volcanic tuff filling the joints in a certain limestone hill, and why are those threads always nearly horizontal? You and Hans say they are not organic fossils, and I accept your conclusion. Rob says that there is no copper in that rock, detectable with his equipment, except within a few millimeters of the green threads. I say nothing except that I have never seen such a thing before. Under the circumstances, I fail to understand where you get the idea that I think there is a city built by the people who stocked this world thirty million years ago buried under that hill. I know I said 'city' when I first saw it, and I still think I was justified in the opinion; I have now seen evidence which causes me to admit that the vegetation pattern was not caused by artificial structures, and I dismiss the original hypothesis. I still want to dig there, and in accordance with Rob's agreement I am going to dig there, with the assistance of anyone who chooses to help. I know you want to go back to your set of leg bones in the cliff, and have no objection to your doing so. Even I can now see, on the basis of your description, that you are uncovering the fossil of a land animal; and I agree that it is of great importance to get it out intact, if possible. But if I can see the importance and even the nature of your work, why can't you do the same for mine?" The little man was leaning forward and staring intensely into Sulewayo's face by the time he finished this harangue, and Ndomi once more felt a trifle ashamed of himself. Lampert, however, saved him the need of formulating an apology.

"I'm sure Ndomi didn't mean to ridicule your work in any way, Take," he said. "We all realize perfectly that an underground phenomenon which cannot be explained at sight either by geology, paleontology or archaeology is something which requires investigation. I imagine that the best plan will be for String and me to go with you tomorrow, while the others continue their stone-cutting. Hans, just how far along are you, anyway?"

The older paleontologist thought for a moment.

"We don't really know," he said at last. "Of course, we aren't trying to get the individual bones completely free of the matrix; that will take somebody months or years. We're uncovering just enough to determine the extent of the specimen, so we can take it all out in one block—or more, of course, if it's too big. So far we can only guess at how big it is. We've uncovered with certainty two feet, and gone about half a meter along one of the attached legs. They seem to be extending straight back into the cliff, so in effect we're cutting a tunnel beside the thing. Assuming it had two main leg sections, as most of the present animals on both Earth and Viridis appear to have, we're about halfway between knee and hip joint. Of course, it might turn out to be the Viridian equivalent of a horse or chicken. In that case, we're about half way between ankle and knee. We certainly have several feet yet to penetrate before we can outline the whole block, assuming that the specimen is essentially complete. Several days, I would guess."

"Can you use any sort of power apparatus for any of your cuts?"

"I don't like to, on general principles, but—yes, we could, with actually very little risk. If you have some sort of rock saw whose cutting part can get fine control, I'd be willing to use it for parts of the tunnel away from the actual specimen."

"I have. We'll take you up there first thing in the morning, and I'll go down with you and show you how to use it before going on with Take and String."

"Who holds the 'copter in place while you climb down the ladder, give your lesson and come back?" asked the guide.

"Hmph. I forgot about that. All right, I'll break out the machinery and give the lesson right now." He got up and strode to the helicopter. McLaughlin covered him from the fence to the aircraft, but nothing dangerous appeared. The geophysicist disappeared inside, and returned a moment later with a

compact metal case under his arm. The guide holstered his weapon as the gate in the fence closed once more....

Actually, the Felodon was miles downstream. It had spent the day in its chosen lair, apparently indifferent to the doings of the men a few hundred yards away. With the coming of darkness—real darkness this time, for the rain clouds cut off both the moonlight and the night glow from the upper atmosphere—it had emerged, hunted, killed and fed as before, apparently unhampered by the lack of light. By midnight it was back in the same lair, paunch distended, as close to sleep as its coldblooded kind ever came.

VI

The rain was still falling when the clouds lightened once more to the rising sun. Lampert was getting used to navigating the canyon by radar, and was an excellent pilot anyway; so he did not have too much trouble in locating the shelf where Sulewayo and Krendall had been working. Getting the men down to it was not particularly difficult, though rather nerve-racking. Krendall went first, unburdened except for his personal equipment. Then he steadied the ladder for Sulewayo who had the cutter strapped across his shoulders. The steadying hand was needed. Climbing down a rope ladder when loaded "top-heavy" can be an extremely awkward bit of activity. Had the pilot above been any less capable, it would probably have been impossible.

The ledge was wet, but fortunately not particularly slippery. The men set their equipment on the ground at the point where their cut entered the crack in the cliff, and without delay set to work. The tunnel was deep enough now to shelter the one actually cutting from the rain, so at first they took turns at this operation.

The cutting machine Lampert had provided was a sort of diamond-toothed chain saw capable of a two-meter extension. Ordinarily it was not the sort of thing a paleontologist would consider using so close to a specimen; but the men were fairly sure by now of the general extent of the thing they were uncovering. Even so, they used the saw only on the side of their tunnel away from the visible remains. They speedily widened the passage enough to

permit them both to get inside and work on the face of the exposed material; but they still used hand tools whenever there was any suspicion that a bone might be about to appear. Work proceeded several times as fast as it had the day before.

They tried cutting another tunnel on the opposite side of the fossil, but this proved rather awkward. The creature was close to this side of the crack, and they had to cut limestone as well as the softer tuff. The saw proved capable of handling this—it would have handled granite without trouble—but went a little more slowly. Eventually, however, the two men were working on opposite sides of the fossil, each in a tunnel extending some two meters into the cliff face.

Half a day's work uncovered the leg bones sufficiently to show that Krendall's first idea had been right. There were only the two major joints, each a trifle shorter than the corresponding parts of the human skeleton. The lower leg was single rather than double, however; knee and ankle both consisted of ball-and-socket joints; and with this fact determined the men paused for thought.

"Now why," mused Krendall aloud, "should any sort of creature need that articulation?"

"Could that foot be a hand instead?" asked Sulewayo.

Of course, questions like that should have awaited the results of detailed examination in a laboratory. Equally of course, the two men proceeded to clear one of the "feet" a little more thoroughly in order to find out for themselves. The answer was not helpful, though.

"He might have picked up a twig with it, but he couldn't have held it any more tightly than I can in my toes," was Krendall's verdict. "It's a bigger and flatter foot than ours. But it's a foot—nothing more."

"Maybe a swimming organ on the side?" suggested Sulewayo cautiously.

"Seems doubtful. If that joint evolved for such a purpose, I should think there'd be a corresponding modification in the foot bones, too—say a flattening such as you see in the paddles of some of the Mesozoic sea reptiles of Earth."

"Reasonable."

"But not necessarily right. That I admit. Anything else strike you?"

"Yes, though it makes the joints still more unbelievable."

"What?"

"The foot itself. Unless some rather remarkable distortion has occurred, it had both longitudinal and transverse arches, like yours and mine—which suggests strongly that this thing's ancestors had been walking erect on two legs for some hundreds of thousands of generations." Krendall raised his eyebrows at this, and silently examined the bony structure before them for several minutes.

"I—hadn't—spotted—that," he said slowly. He looked in silence for several more seconds. Then the two men, moved by a single thought, went to the other end of the exposed leg and began to clear the hip joint and pelvic region. They worked almost in silence, understanding each other perfectly, like an experienced surgical team; and gradually the equivalent of a pelvic girdle and lower end of a spinal column were cleared sufficiently to show their general nature.

It was at this point that the helicopter returned; but neither man noticed the fact until McLaughlin had called several times from the open ladder hatch. They climbed silently and thoughtfully up to the flyer; but Mitsuitei's first question started the talk flowing.

It did not end for a long, long time.

Krendall, with difficulty, held interruptions of his more volatile companion.

"There can be only the slightest doubt that this thing we're uncovering walked erect on two legs," he reported. "The feet; the way the pelvis is modified to *support* internal organs; the fusing of the lowest vertebrae with the pelvic girdle to form a weight carrying foundation—they all point the same way. The only thing hard to understand is the knee and ankle joints. If we had them, it would be virtually impossible for us to hold our legs rigid. Perhaps some really remarkable musculature—"

"Or a cartilage structure which has not been preserved," cut in Sulewayo.

"Or some such thing as that, would explain it. I don't know. The creature is good for several Ph.D. theses just as it lies—and probably an equal number of nervous collapses when we get it out."

"I find myself strongly desirous of seeing its skull," remarked Lampert. Sulewayo glanced at him sharply.

"You, too?" asked the young paleontologist. "I was hoping I was the only one crazy enough to have thought of that." Mitsuitei smiled openly, an almost unheard-of act for him. He said nothing for a moment, but everyone saw him; and even McLaughlin understood the thought. After a sufficiently long pause, he asked a question.

"Have you uncovered enough of this creature's structure to guess at any evolutionary connection—or lack of it—with the amphibids we already know on this world?"

"I'd hate to take any oaths," replied Krendall. "The legs, which we've seen most of, are different in detail; but they at least correspond in general with what we find here. The only really significant point there would be the single shin-bone. In that it resembles Viridian land life in general—these animals don't have the separate tibia and fibula characteristic of the usual run of Earthly land vertebrates. It really proves nothing about what we're all thinking, of course."

"I am tempted to work with you gentlemen tomorrow," muttered the archaeologist.

"Why? Didn't your investigation pan out?"

"It is harder for me to say than for you, so far. To dig a pit, big enough not only to work in but to cover a useful amount of ground, in a driving rain, is quite a job even with Rob's machines—which I would never use were I not sure that there is nothing of importance above the limestone level. I have gotten down to the rock over an area three meters square, which is very good going; but I shall undoubtedly find the pit full of water tomorrow, as we have not yet improvised a really satisfactory drainage system. I cannot—or at least will not—use machines inside the crack in the limestone; so it will be some time before I get down to our mysterious green threads."

"Then it would seem that the best we can do is go on as we have," said Lampert. "The only change might be if one more man were to help at Take's

dig. But I don't suppose either Hans or Ndomi would care to leave his own job at the moment, and actually there's not much more to do at the hill which can be done by anyone but Take himself. I'll continue to help him as long as it's a question of moving mud, but after that he'll have to do his own sifting. String is automatically on guard duty at the hill, so there's not much change we can make. Though I must say I haven't seen anything dangerous yet, in that jungle."

"Those animals are like crows," remarked the guide. "We used to have 'em on the farm, back on Earth. They'd be all over a freshly planted field, while no one was around. Come out yelling—they don't move; come out with a gun, and they're gone—unless you'd happened to forget to load it; then they sat and laughed at you. If you're suggesting, Doctor, that I should relax the guard duty and lend a hand with digging, I veto the idea—and not because I'm afraid of getting my hands dirty."

"I won't say I didn't have some such thought, but I accept your ruling," smiled Lampert. There was silence for a moment; then Krendall reverted to the earlier subject.

"You know," he said, "if this thing we've found does turn out to have been intelligent, it will hardly solve any of the existing problems about Viridis."

"Why not?" asked Sulewayo in some surprise.

"We still won't know whether it's native to the planet or not, unless we can establish a relatively complete evolutionary sequence leading to this form. If we do that, the question of speed of evolution here gets worse than ever; if we don't no one will be sure whether or not we ought to look for buried spaceports or send out expeditions to find the planet they might have come from."

"The latter would be something of a waste of time," remarked McLaughlin. "Hunting one planet in the galaxy is like hunting one log of wood on Viridis." No one contradicted this. All had seen the galactic star clouds from outside planetary atmosphere.

"It seems to me, speaking as an amateur in your fields, gentlemen," said Mitsuitei, "that the mere discovery of an intelligent creature in the Viridian fossil deposits would, on the basis of our present knowledge of the mechanisms of evolution, strongly support the idea that this world was stocked from others. I realize that our knowledge *may* not be sufficient to justify us in that conclusion. But it is *certainly* not great enough to justify any other."

"You seem to have something there, Take," admitted Krendall. "If this thing does turn out to have room for a brain in its skull, I suppose the next ten conventions of the Interstellar Archaeological Society, or whatever you call it, will be meeting at Emeraude."

"I shouldn't be at all surprised. So far, my profession and yours have not overlapped, due to a considerable factor of difference in the time spans covered. But it is just possible that we would be holding joint meetings, in the event you describe."

"This meeting is changing from discussion to speculation," Lampert said drily. "I would be the last to decry the value of imagination; but actually we are as likely to face the need for entirely new hypotheses as the result of our work here, as to find support for any now in existence. I can speculate with the best of you, but for goodness sake let's not take any speculation too seriously. I don't *really* believe that some big-headed descendants of Ndomi's fossil are listening in on me right now!"

Even Sulewayo admitted that this was rather unlikely, and the conversation turned to other matters until darkness fell.

No one had trouble sleeping. The loud drumming of the rain on the metal roof meant nothing to field workers with their experience. If anything, the sound was soothing, giving a perpetual reminder that there was a roof. Such protection is not always available, in that line of work....



The Felodon seemed to have lost its traveling propensity. Once more it went out into the utter darkness solely to get a meal. It accomplished this as quickly as ever, though its eyes must have been useless and the hiss and rumble of falling water drowned and buried any sounds which would have been useful in tracking. Back in the same lair, full-fed, it drowsed once more.

VII

Mitsuitei had been almost right in his prediction that the pit would be full of water. Only the fact that the land sloped a trifle—they were not right on top of the little hill—had saved it. As it was, several feet of water were in the bottom, and a good deal of mud had washed in from the two sides facing the

edges of the crack. The other two, much better braced by deep-reaching roots, had held firm.

After some thought, Lampert used the little robot again. He started it at the bottom of the pit on the downhill side and drove almost horizontally toward the river. The two hundred meters of "neck" permitted the mole to emerge from the slope farther down. When it was withdrawn, a small drain hole was obtained. Several more of these were drilled, and the pit lost its water fairly rapidly.

There was still the problem of getting into the crack itself, which of course would involve digging below the level of the drain holes. Lampert, using the same excavator which had made the pit itself, finally provided a fair solution by digging a set of ditches around the larger hole; and since the opening itself was quite well protected by over-hanging trees, Mitsuitei had only drainage from the surrounding soil to contend with.

Two hours after arriving, therefore, he had a relatively clear working space. The bottom of the pit was limestone, exposed by the complete removal of the overlying soil, some three meters square. Across it ran the crack, a trifle less than a meter wide, still packed with dirt. Everything was muddy—limestone, projecting roots, and Mitsuitei himself. A slender log with branches cut to ten-centimeter stubs leaned against one corner, forming a rough ladder and giving entrance and egress to and from the site.

The machinery which had done the original digging was at one side. Mitsuitei did not expect to need it again. He was now equipped with a hand shovel, and seemed about to use it. Lampert, standing at the edge of the pit, felt the incongruity, but managed not to laugh.

"Are you sure there's nothing I can do down there with you?" he asked.

"I'm afraid not. From now on I want every bit of dirt to pass under my own eyes."

"Are you going to try to throw it all up here as you finish?"

"No. That's the purpose of the extra pit area down here. I can get a long way down the joint, simply heaping the material on the rock. It's damp enough to pile quite steeply, too."

"How far down do you think you can get? The crack's rather narrow to work in, and you have three and a half meters to go before you hit tuff. That's

going to be rough shoveling. I still think you could use the machine safely for a little way further, at least."

"No doubt I could, but I'm not going to. There's one thing I might use, though. If you have another of those saws, such as the bonemen are using up on the cliff, I could widen this crack as I go—cut steps, in fact, to help get the mud up to this level when I'm further down."

"That's a good thought, but I don't have any other. If you really get far enough down to need it, though, I could fly up to get it. They were going to shift over to hand labor anyway."

"All right. Of course, it will be some time before I get that deep anyway; maybe I won't need it today." He bent to his work.

"But what do I do?" asked Lampert. "I can't go off to attend to my own projects, because String has to stay here to guard you. I can't get to the site where the others are working because I can't land there. I can't sit in the helicopter and twiddle my thumbs because I'll go crazy before the day is over." Mitsuitei straightened once more, and thought briefly.

"Is there nothing in the geophysical line you could do within sight of this pit?" he asked finally. "The saw and digging machine are not the only apparatus you brought."

"That's true. I brought some seismic gear, though I didn't plan to use it quite like this. I might map the formations under this hill. The information will be usable, I should think, and the joints will give quite a calibrating job. It will keep me busy, anyway."

"Just a minute!" Mitsuitei looked a trifle perturbed. "Does that mean you're going to set off explosives around here? I want the sides of this pit held up by something better than roots, if you do."

Lampert chuckled. "No explosives," he said. "This is a nice little gadget with a robot like the core sampler. It puts out waves of any type desired from any depth down to two hundred fifty meters—a sort of subterranean sonar. You'll never know it's working. The wave amplitude isn't enough to feel." He turned toward the helicopter on the river bank below, and was starting to

walk toward it when McLaughlin interrupted. The guide had heard the conversation, and his question was purely rhetorical.

"You weren't planning to walk down to the flyer alone, were you, Doctor?"

"Well, yes, as a matter of fact. After all, I won't be working; I can keep my eyes open as I go. You can see me for the greater part of the journey from here, too."

Rather to his surprise the guide approved this argument, after a moment's thought.

"All right. But please keep your gun in your hand as well as your head on a swivel. I'd prefer to have Dr. Mitsuitei come down with us so we could stay together, but I know how he'd react to the interruption, and I realize you're not a kid. Just be careful."

Lampert promised; and the guide's manner had impressed him to the point where he was almost afraid to make the return journey, after reaching the flyer and packing his new equipment. He was rather surprised to get back to the site without being attacked, and McLaughlin's very evident relief at seeing him did nothing to ease his feelings.

He began to set up the machinery. This consisted of an assembly very similar to the drilling mole—a small delving robot drawing a slender tail behind it, the tail wound on a drum which surrounded the control unit. A dozen smaller cylinders reposed in attached clips.

"The attached borer," Lampert explained to the guide, "goes down to any depth I set, up to two hundred fifty meters. It can produce any of the three normal types of earthquake wave, singly or in any combination, with sufficient intensity to be detected at a range of over two kilometers in reasonably well-conducting rock. The small cylinders are detectors, equipped not only to receive and analyze the wave coming through the ground but to measure electronically their location with respect to each other and the main station. I can use as many of them as I please, up to the full dozen; but they can be planted only a little way below the surface. There exists equipment for getting readings at depths comparable to that of the transmitter, but I don't have it. As it stands, by spotting the receivers carefully I can get a pretty good picture of the formations for a radius of a

kilometer and a depth even greater with ten minutes measuring—and ten hours computing."

"How far out do you plan to place these receivers?" the guide asked pointedly.

"Well—I hadn't made a detailed plan of that. I'd rather like to have them in radiating lines of three, the lines spreading about fifteen degrees, and the individual cylinders about two hundred meters apart."

"And just how were you going to place them? I gather that someone has to walk the best part of four kilometers—or do these things fly, in addition to their other abilities?"

"Er—someone walks. I thought perhaps, since you don't like the idea of my going alone through the jungle, that I might stand guard over Take in the pit while you set them out."

"Hmm." The guide did not explode, to Lampert's relief. It had not occurred to the scientist that the job of wandering around a hole in the ground waiting for animals which never came might get a little boring to a man of McLaughlin's background. "Let's go over first and see how Dr. Mitsuitei is getting along. I guess you could stand over him with a gun for half an hour. Of course, the cover runs dangerously close to the pit. Maybe we'd better burn it off to a safer distance—still, I guess that won't be necessary. You can stand out here where it's relatively clear, and see all the approaches to the pit. Something might jump in without your having time to hit it, and you'd at least see it and could get there fast enough to do any shooting necessary."

They approached the hole and looked in. Mitsuitei was working busily. A fair quantity of earth lay spread on the rock, and some two thirds of the length of the crack had been excavated to a depth of perhaps a quarter of a meter. The geophysicist attracted the little man's attention and told him of the plan; Mitsuitei nodded and bent once more to his work.

The Felodon was becoming restless. It could hardly be hungry as yet; but it was on its feet, snarling silently as it had when the helicopter first entered its ken. For perhaps a minute it stood; then, with the same air of determination it had shown days before and scores of kilometers away, it began to thread its way through the underbrush toward the river—and the digging site.

"I'll stand where you suggested, and never take my eyes off the pit," Lampert promised.

"Then I'll come back to find you missing," replied the guide. "You're guarding yourself too, remember. Don't keep your eyes on anything. Keep them moving."

He finished distributing the little cylinders in the various pockets of his outer clothing, and moved off in the direction Lampert had indicated. He looked back frequently, but each time saw the scientist alert. When the underbrush finally cut off the view, he refused to worry too much.

Actually, McLaughlin had gone to considerable pains to make the jungles of Viridis sound more dangerous than they really are. His conscious motive was to make the inexperienced members of the party alert enough for their own safety. It was quite true that a man could be killed in quite a variety of ways in those rain forests. There was a distinct possibility, however, that he also wanted to impress them with the importance of his services.

He did not, therefore, suffer much from anxiety during his walk, though on the other hand he wasted no time. He had, of course, only a rough idea of the distance he had traveled, though he was able to keep his direction with a small impulse-compass tuned to the seismic apparatus and forming part of its regular equipment.

He dropped three of the cylinders at the required intervals, as nearly as he could guess, forcing each a little way into the ground as Lampert had shown him; then he turned at right angles, walked what he hoped was the right distance and started back toward the site, planting equipment as he went. Out again, in again; and the last of the dozen tubes was in the ground.

Mitsuitei's shovel scraped deeper.

Lampert, glancing up and around every few seconds, made minute adjustments to the controls of his seismic apparatus. Its little mole robot had started on its downward trip.

The Felodon lurked thirty yards from the point where Lampert was standing, protected from his sight by the undergrowth and by one of the piles of dirt thrown up by the machine which had dug the pit. It seemed to be looking

through the soil at the spot where the man was. The snarl was still on its face, but no muscle moved in its long body. It had been there for minutes without moving; it had frozen similarly when McLaughlin had passed it on his way out. Now it simply stood and waited.

On a cliffside kilometers away, Ndomi Sulewayo gave utterance to the first profanity Krendall had ever heard him use. They were on opposite sides of the block containing the fossil, so neither could see the other. Krendall, naturally, asked what was wrong.

"Don't tell me a bug got through one of these suits!"

"Worse, if possible. I told you this foreleg—" both had been carefully avoiding the use of such words as "arms"—"was sticking out sideways, so that I was afraid we might have cut off part of it in digging the tunnel."

Krendall nodded. "I remember. Did we?"

"I don't know."

"Eh? How come? I should think there'd be no doubt, one way or the other, if you have that much of the limb clear."

"Well, I haven't. I got as far as the bone goes—and right there I run out of tuff and into the limestone. If there's anything more, it's in an entirely different kind of rock, which is a trifle unlikely; but I'm going to have to check the blocks we cut from this part of the tunnel in order to make sure, and I don't look forward to the job at all." Krendall, properly sympathetic, came around to Sulewayo's side to look, and agreed that the search was necessary. The bone the younger man had been clearing ended in a joint of the type they had come to regard as typical of the creature's limbs; and this had occurred almost exactly at the surface they had left when first outlining the block with the saw.

Sulewayo, with a grunt of disgust, dropped his tools and went out into the rain, where the blocks cut from the cliff had been piled; Krendall, nobly sacrificing his personal inclinations, went along with him.

The search lasted for a long time; for a long time, in fact, after it became evident that it was going to be useless, for the chance of a perfect specimen is not easily thrown away. Finally, however, Krendall straightened up with a sigh.

"I guess we'll have to be satisfied with a restoration on one side," he said wearily. "I hope someone fifty years from now doesn't find another and discover that it's a sort of vertebrate fiddler crab, with one fore-limb ending in a paw or claw something like five times the size of the one on the other."

Sulewayo gave a gloomy assent, and the two went back to work in their respective tunnels.

Lampert saw McLaughlin the instant the underbrush made it possible, a fact which the guide later admitted was to the scientist's credit. He had, of course, been eagerly awaiting that return, for the transmitter was down to its first set depth and awaiting only the word that all receivers were in place. He called eagerly the moment the guide came within earshot.

"Everything down?" McLaughlin nodded.

"Everything down, as nearly as I could tell the way you said. How long will the readings take?"

"Only a few minutes. I'll take a couple of calibration shots from ten, fifteen and twenty meters' depth; then ones at fifty, a hundred and so on down as far as the mole will go. The shooting takes practically no time. It's the drilling that will hold us up."

"What then?"

"Well," Lampert smiled, "after that the usual procedure is to pick up the receivers and place them in a similar pattern in a new direction. If the field crew doesn't go on strike, we take the whole circle about the transmitter."

"I was afraid of that," grunted McLaughlin, as he stopped by the machine. "Well, let's go." The two men bent over the controls in a silence broken only by the scraping of Mitsuitei's shovel a dozen meters away. Lampert pressed his shot button, and a light on the panel flashed white momentarily. Below their feet, unfelt, the pulse of sound energy raced outward, echoing from the walls of deepstriking joints, from the boundaries between rocks of differing densities or elastic constants, from the walls of caverns deep in the

limestone; some tiny portion of the energy from time to time encountering and affecting one of the tiny receivers McLaughlin had buried.

As each receiver gathered its bit of data, it retransmitted the information to the master unit; and everything was recorded on a single sheet as the milliseconds sped by. Long before a full second had passed, the first of the pulses had damped out as heat energy, and enough had been transmitted for the machine to obtain an adequate averaging record. The light blinked out again. Lampert nodded in satisfaction, and sent the mole downward once more.

"Look, good. Now the next set," he remarked.

As that pulse of seismic energy went forth, the Felodon rose to its full height, almost showing itself over the pile of dirt which was now its sole protection from the view of the men. The snarl on its face seemed to grow fiercer, if that were possible. For just an instant it seemed torn by conflicting desires. But that was for just an instant; any tendency to flee was smothered before it could take full form. There were two men now to worry about, and correspondingly less chance for the opportunity it had been awaiting. But the opportunity came. For just a moment the guide looked down at the panel which was absorbing Lampert's full attention. In that moment a green-and-lavender streak flowed over the heap of soil in a single leap and vanished into the pit. It must have been timed and guided by the mysterious sense McLaughlin had mentioned. It could see none of the men when it leaped, yet it timed the act for the moment none were looking, and landed directly on Mitsuitei.

The little archaeologist never knew what hit him. He died without a sound, and the killer, as though nothing lived anywhere in the neighborhood, settled down to its meal.

In this it must have been disappointed. The chemicals in the clothing designed to repel Viridian insects were equally obnoxious to the carnivore, and it made no serious attempt to get through them. However, not all of the body was protected in this way....

A second pulse went from the buried transmitter, and then a third, each from a point a few meters deeper than the last. Lampert's attention, of course, was centered on his controls. McLaughlin's eyes were once more sweeping restlessly over the surrounding landscape. Both heard the sounds coming from the pit, but neither interpreted them as anything more than the scraping of Mitsuitei's shovel. Neither, of course, considered them consciously. Their attention was finally attracted by something decidedly more noticeable.

The Felodon did not—or could not?—remain at its meal for more than a few moments. Its apparent indifference to the other men changed once more to what seemed like an internal struggle. An observer would have been sure, up to now, that it was using its peculiar sense to avoid the sight of men with guns; but that hypothesis failed now.

As Lampert started the mole robot downward once more, the Felodon leaped out of the pit toward the two men—regardless of the fact that McLaughlin was facing toward it.

VIII

McLaughlin saw the fanged head emerge, and his reflexes took over instantly. A streak of flame passed beside the leaping carnivore, exploding into a white-hot blossom of blazing gas as it contacted the pile of dirt on the far side of the pit. The guide ducked and rolled frantically sideward as another spring carried the creature toward him. Claws raked the air past his shoulder, and he fired again before the roll was complete and without any sort of aim.

Men and beast alike were spattered with white-hot droplets of metal from the seismic recorder as the second shot caught it squarely; and this seemed to be enough for the carnivore. Its next leap was away from the men instead of toward them. A geyser of steam and mud erupted beside it as Lampert finally got his weapon into action, and before the vapor had been beaten down once more by the rain the animal was out of sight behind the undergrowth. Both men sent several shots in the direction of the crackling bushes, but accomplished nothing except the felling of a tree or two and the starting of a bonfire which failed to make any headway against the rain.

Convinced that the Felodon had gone, the men ran to the pit. Lampert did not even take time out to glance at the wreckage of his equipment. There was just enough distance to cover to let each one realize that he had no idea how long the carnivore had been inside, and what the "scraping" sound might have been. Both slowed down as they approached the edge, not relishing what they expected to see. But this did not prove to be what they had expected. McLaughlin's face, already grim, turned gray as he saw that his first shot had not merely missed the animal at which it was aimed.

The bolt had struck the pile of dirt which had been left by the digging machinery at the far lip of the pit, and scattered most of it to the four winds. Perhaps half a ton had slid back into the hole from which it had originally been removed. There was no telling, from above, what the Felodon had done to Mitsuitei. The upper half of the archaeologist's body was buried completely, and the rest so liberally sprinkled with dirt that it was not at once identifiable.

The guide, using language strange even to the widely-traveled Lampert, leaped the three meters downward without bothering to use the ladder, seized a projecting leg and tried to draw the little man clear of the soil. Lampert, equally aware of the possible value of time but feeling that he would do little good with a broken leg, made the descent in the normal manner.

By the time he reached the bottom, McLaughlin had succeeded in dragging Mitsuitei almost completely clear. Lampert started forward to clear the mud from the still hidden face; then he stopped, and his stomach abruptly heaved, as he realized that the face was not hidden.

It was gone.

Mitsuitei had removed the head-gear and gloves from his protective suit for the normal reason—to see and manipulate better. The exposed head and hands had formed the Felodon's hasty meal.

The paleontologists saw the helicopter approaching this time, for they were working outside the tunnel. Between them on the ledge lay a block of stone some five feet long, two high and four wide—over two tons of material, all



Sulewayo was first up the ladder, unburdened this time. They expected to have further use for the cutter. He noted that Lampert was alone in the machine, and promptly asked the question the geophysicist had been dreading.

"Where's Take? We've found something for him!"

"I'm afraid he won't appreciate it. He was killed a couple of hours ago by a Felodon." The news silenced even Sulewayo, and the expression on his junior's face actually startled Krendall when he climbed through the hatch.

"Ndomi! What in—" Lampert cut in with the same news he had given a moment before. Krendall reacted similarly; then slowly lowered himself

into a seat.

He did not ask for details. Both men could see that this was not the time to put such a question to the pilot, though neither realized then the personal responsibility that Lampert felt over the incident. Krendall pulled a small fragment of tuff from his pocket, and looked at it thoughtfully.

Nothing further was said until the helicopter landed once more on the river near the "city." McLaughlin and the bundle which held what was left of Takehiko Mitsuitei were waiting on the bank, and were loaded aboard without a sound.

"It's early. We'll take him back to Emeraude tonight, and come back for your work tomorrow," Lampert said, and lifted into the air without waiting for agreement.

"All right," replied Krendall, "as long as we come back. I don't think he'd have wanted us to stop. I'm going to find out about those green threads of his, too." Lampert nodded in approval. He had already formed a similar determination. For half an hour they flew on in silence.

The Felodon, half submerged in swamp water a kilometer downstream from the hill, heard the helicopter hum overhead. It seemed totally disinterested. For just a moment its fanged head pointed upward, then settled back again. There was a burn under its jaw, which had been inflicted by metal spraying from the ruined seismic apparatus. It was more comfortable to keep it under water....

"What was it you found, Ndomi, that you thought Take would want to see?" Lampert broke the long silence.

"It was when we were undercutting to get the block out of the tunnel," Sulewayo answered. "It's just some more of his green threads, in the tuff below the fossil. I brought a chunk of the rock showing them—here." Lampert nodded without taking his main attention from flying.

"Maybe that fossil of yours was intelligent after all, then," he said. "It seems to have died under very similar circumstances to Take—just above a set of those green threads. Maybe it was a member of a party like ours."

"Maybe. It certainly walked erect. The whole body structure shows that. If its brain were large enough and it had some sort of manipulating appendage

I'd say it was virtually human—in capacity, that is. It was more of an amphibian anatomically."

"You have the block out in the open. Haven't you been able to study the head and limbs?"

"No, damn it." Krendall took over from his junior. "That was the big disappointment of the whole find. The specimen seems to be perfect except for missing skull and hands. Not a trace of either."

The helicopter wavered slightly in its path, then steadied as Lampert forced his attention back to his job. No one said anything for a long time, but everyone was thinking.

Someone else was thinking too, but wasn't keeping his thoughts to himself. They were being spoken, and virtually dripped with the thinker's fury.

"You sloppy, lazy parasites! I don't mind being stuck with a job and a deadline, even if it's a report that's due in only fifty years and needs about two hundred for a real conclusion. I'd sooner do it all myself than have some of you loose thinkers butting in! But if I'm to give my whole attention to it so I can produce something that won't be laughed at from here to the Magellanic Clouds, how about some of you watching what goes on on this planet? I didn't know those creatures were poking around until they began cutting sensor lines! Thirty, the protective life we've bred on the surface was your idea; why didn't you put it to work?"

The answering words tried to be soothing.

"Thirty was working—"

"Dreaming, you mean!"

"But I put one of the guardians on the job for him; it stopped the digging, didn't it?"

"Sure, after a lot of damage had been done. Do *you* want to come out in the open and repair my wiring—with those space travelers poking around? If they find you at it, the Council won't just laugh at us; they'll excommunicate

us and let this new species of intelligence clean us out. Why did it take so long for the guardian to do anything?"

"Well, I—"

"Well, you were dreaming too, weren't you. Blast it, you're here to do constructive thinking, not just to entertain yourself. Haven't you any self-respect? They actually dug out Sixteen!"

"What difference will that make? He's been dead too long to mind it himself, and anyway his brain and sensor connections were decently burned."

"But they wouldn't have had to dig much deeper to get someone who's *not* dead, would they, Ninety-Five, my young friend? I suppose when they cut one or two of *your* sensors you decided it was time to do something. Don't interrupt! I'm talking! This planet is supposed to be a quiet place where people can expect to spend a decent number of centuries at a time thinking, *without* being disturbed. If you're too young or too lazy or just too stupid to do any real thinking yourself, at least you can devote a little time from your casual amusements to making sure that other people *can*. Shut up! You'll do some thinking now or find yourself in real trouble! Here's a problem for you to solve, and see that you solve it!

"You will get my sensors repaired, making sure not only that you're not caught at the job by these space travelers but also that they don't realize it's been done. In other words, don't just neatly fill in the hole they made after you've finished, so they can't help knowing they're not the only intelligences on this world. I don't know when they'll be back any better than you do, so you'll have to guess at your own time limit. You can booby-trap your canyon with landslides or anything else to keep yourself from being dug out, but if you fail in either problem and either of us looks likely to be found I personally guarantee you'll be found in the same shape as Sixteen. Now get to work, and let me think. If you think you can get help or sympathy from anyone else on the planet, good luck to you."

A wave of agreement spread along the countless miles of sensor wiring that extended through Viridis' crust, but Twenty-Five didn't feel or hear it. He had already taken the myriad of tendrils that terminated his arms away from the mosaic console that formed the end of the vast bundles of greenish

threads coming through the walls of his cave, and had settled back in his lounging chair. That report—only fifty years to have it thought out—his full attention went back to it.

*** END OF THE PROJECT GUTENBERG EBOOK THE GREEN
WORLD ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.